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Proportionality as a promising method to enhancing functional inference by co-expression analysis

Scarlet Sung

Abstract

Co-expression analysis is widely used for gene function inference. Conventional correlation calculation methods in co-expression analysis, however, rely on normalization strategies that have been found to introduce spurious correlation effects. A proposed explanation suggests that expression data is compositional and should thus be analyzed with consideration of the relative relationships of its components, such as genes, rather than using conventional correlation measures. In this report, different workflows incorporating different combinations of correlation calculations, proportionality measures, and data transformation methods are compared. The correlation distributions produced by the different workflows were investigated to identify signs of compositional effects. Correlation-based co-expression analysis methods are systematically compared to proportionality-based methods on their ability to accurately recover established functional and structural relationships between genes. This report shows that proportionality-based methods more effectively recover functional and structural relationships between genes in co-expression analysis, and thus demonstrate promise in improving the robustness of co-expression analysis in bacterial species. The resulting potential improvements in gene function prediction robustness could further expedite the identification of novel gene candidates for biological, agricultural, and medical applications.

Introduction

Advancements in high-throughput sequencing technologies have allowed for the time- and cost-efficient generation of an abundance of datasets, allowing for rapid advancements in the characterization of genomes and transcriptomes. RNA-seq is one of such technologies that quantifies gene expression[1]. Since gene expression is known to be dependent on a variety of biological and conditional factors, RNA-seq experiments and subsequent downstream analysis are often used for functional inference[2]. Specifically, given that many loci are expressed differently under different time, strains, and stimuli, expression patterns associated with specific conditions or co-expression between genes can offer insights into gene function[1] and expedite the identification of candidate genes for further experimentation[3]. Gene function inference with RNA-seq has led to important insights into fruit production[4], pathogenesis[5], transcription regulation[6], and cancer biomarkers[7], as examples amongst many other applications[8].

Coordinated expression patterns are well-observed amongst genes involved in common biological processes and have been shown to be a reliable indicator of co-regulation[9]. Therefore, it is possible to use variations of the Guilt-By-Association (GBA) principle, which states that co-expressed genes usually participate in the same or similar cellular functions[10,11]. Co-expression analysis incorporates this idea to systematically predict the function of genes or gene clusters, inferring from highly correlated genes of known functions[12,13]. This is done by calculating similarity-measures between each pair of genes of the dataset, usually with a correlation-measure such as Pearson correlation coefficient (PCC) or Spearman correlation, to identify genes of similar functions or construct networks illustrating relationships between biological pathways. Prior to the calculation of the similarity matrix, however, RNA-seq raw counts conventionally undergo several levels of transformation and normalization to

facilitate accurate comparisons between samples[14]. At the time of this report, as far as the author is aware, Weighted Gene Co-expression Network Analysis (WGCNA)[15] remains the most widely used and accepted tool for co-expression analysis[16–18]. WGCNA uses PCC alongside normalization techniques, illustrating the widespread acceptance of this pipeline.

While co-expression analysis has been shown to be effective in functional inference[6-8], skewed normal distributions have been identified after the normalization and log-transformation of raw RNA-seq counts during the development of software tool COCOLOCUS[19]. COCOLOCUS is an R-package developed in-house that aims to infer functions and refine boundaries of biosynthetic gene clusters (BGCs) using gene-expression data and subsequent co-expression analysis. In the development of its co-expression analysis workflow, however, it has been observed that the distribution of correlation coefficients amongst genes with higher expression tends to skew positive. This phenomenon, also known as a mean-correlation relationship, was also described in literature[20], and does not align with the normality assumption required for PCC calculations. The COCOLOCUS workflow thus provides the options to transform expression using mean residuals or first principal component (PC1) residuals to normalize correlations[21], as well as applying spatial quantile normalization (SpQN)[20] alongside PCC. It could be argued, however, that the abundance of data transformations may introduce artifactual biases to the data in and of itself. In addition, these transformations do not always sufficiently adjust for the skewed distributions (Figure 2). Addressing normalization strategies and potential artifactual biases in co-expression analysis was thus the original motivation for this project.

The observed biases could potentially be explained by the compositional nature of RNA-seq data. Compositional data is a set of non-zero positive values carrying relative information[22] where the sum of its values is arbitrarily defined by the sampling procedure and where the differences between values can only be meaningfully interpreted proportionally[23]. It is suggested that RNA-seq data is compositional due to being limited by a library size defined by the assay technology, and thus the conventional correlation measures simply do not apply[24]. Compositional data analysis (CoDA) has been developed to address this phenomenon. With CoDA, datasets are transformed using log-transformation where the data is scaled with a select reference value. One such transformation is Centered log-ratio (CLR) transformation, a CoDA transformation method that uses the geometric mean of a data vector as the reference[25]. The association between the log-transformed datasets is referred to as proportionality, analogous to correlation as calculated by conventional correlation measures. This method does not assume normality and aims to reflect the relative relationships between samples[26].

Proportionality and CoDA, as it stands, is not widely used in the context of co-expression analysis. Though awareness of the compositional nature of RNA-seq data is growing and some research evaluating its efficacy have shown promise[26–30]. However, the application of log-ratio transformations in CoDA to gene expression data can be challenging due to their discrete nature[31,32], leading to hesitance in using proportionality measures as an alternative. Thus, whether or not a transition to CoDA as a standard for co-expression analysis is necessary remains controversial, and the limited scope of its use so far begs further exploration. Therefore, to improve the robustness of co-expression analysis based functional predictions for applications such as COCOLOCUS, we investigate whether functional inference based on existing co-expression analysis in COCOLOCUS and similar workflows are sufficiently accurate and statistically sound. We evaluate if proportionality-based co-expression analysis methodology improves the detection of co-regulated pathways and prediction of gene function compared to traditional correlation measures alongside commonly used transformation methods. Through this, we explore the extent to which the compositional nature of expression data impacts correlation analysis robustness. The results of this report may provide broader insights into strategies to obtain more accurate gene function predictions.

In this report, we evaluate different gene-expression transformation and similarity measure calculation methods through systematically comparing its predicted top similar gene pairs to known functional associations and operon categorization. We show that proportionality-based similarity measures demonstrate promise as a more effective co-expression analysis method than conventional correlation-based measures. We conclude by discussing the strengths and limitations of this methodology exploration and the wider implications of these results in gene functional inference.

Materials and Methods

Data acquisition

Publicly available RNA-seq datasets were used to develop, test, and evaluate the co-expression analysis methods. The co-expression analysis portion of COCOLOCUS[19] was developed and tested with published *Burkholderia pseudomallei* (*B. pseudomallei*) K96243 gene expression profiling data[33]. For consistency, the *B. pseudomallei* dataset is also used in this project. To further evaluate the performances of the co-expression analysis methods, additional RNA-seq expression datasets were selected from NCBI Gene Expression Omnibus (NCBI GEO)[34,35]. To limit the scope of the project due to time constraints, only bulk RNA-seq data was used, as approaches may differ with single-cell RNA-sequencing (scRNA-seq)[36]. Selected datasets contain at least 10 samples (conditions x replicates)[15], and both genes of at least 10% of gene pairs are annotated to at least one Gene Ontology (GO)-term[37] to ensure meaningful evaluations. Of the list of suitable candidates, datasets were selected to represent a range of species and chosen based on initial availability. Datasets selected were of *Escherichia coli* (*E. coli*) K-12 MG1655[38] and of *Bacillus subtilis* (*B. subtilis*) str.168[39] (Table S1).

Data preparation

All expression datasets were filtered prior to all tested co-expression analysis methods with the function filterLowExpression from the COCOLOCUS R-package (unpublished version) using default parameters. Zero imputation was additionally performed prior to two methods using the function cmultRepl from R-package "zCompositions" (v1.6.0)[40,41]. The Count Zero Multiplicative (CZM) replacement method was selected as recommended for count-based expression data[42], and "pseudocounts" was selected to convert calculated proportions back to counts for consistency across all methods.

Co-expression analysis

To evaluate proportionality-based co-expression analysis methods, two different R-packages "compositions" (v2.0-9)[23,43] and "propr"(v5.1.8)[26] were tested. Zero imputation with zCompositions was performed for each method. CLR was performed with "compositions" with function "clr" to perform CLR and "clrInv" to convert the transformed data back into proportions for interpretability. PCC was then performed to calculate correlation with the "cor" function from the WGCNA R-Package (v1.74)[15]. CLR was also performed with propr which calculates rho as a similarity-measure. Since propr has a built-in zero imputation method, propr without zCompositions zero imputation was also compared. The workflow using R-package "compositions" will hereby be referred to as "CLR", the R-package "propr" alone will be referred to as "propr", and the R-package "propr" with zero imputation with zCompositions will hereby be referred to as "propr + zCompositions".

These methods were compared to the existing COCOLOCUS co-expression analysis workflow representing conventional co-expression analysis methodology. The filtered expression matrix is normalized with trimmed mean of M-values (TMM) and log-transformed with the function logTMM. Then, mean residuals and PC1 residuals were calculated independently from the matrix with function

meanResiduals and removePCs respectively. Correlation matrices were then calculated for each of the logTMM, mean residuals, and PC1 expression matrices, using the function corAndSpQN. This function calculates the correlation matrices with WGCNA::cor, then addresses the mean-correlation relationship bias with the function normalize_correlation from Bioconductor (v3.22)[44] R-package "spqn"(v1.22.0)[20,45]. The resulting correlation matrices for the logTMM, mean residuals, and PC1 residuals methods will hereby be referred to as "Original", "Mean Residuals", and "PC1 Residuals" respectively. Workflows are illustrated in Figure 1.

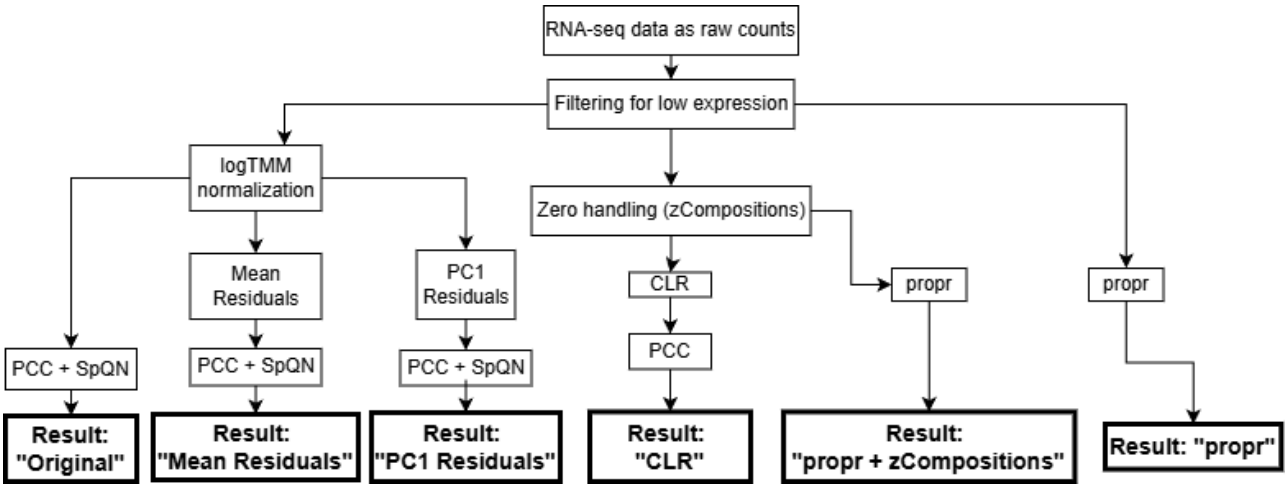

Figure 1. *Flowchart illustrating the overall workflow for all evaluated co-expression analysis methods.*

Data distribution was visualized with density plots of correlations within subsets to preliminarily identify biases. Specifically, within the bottom 10%, middle 10%, top 10% of genes by gene expression, as well as the correlation between the bottom and top 10% of genes. After the similarity matrices were generated, all gene pairs with their corresponding similarity measure were extracted from each matrix for evaluation. Gene pairs for each method were sorted by decreasing correlation.

The function bench::mark from R-package "bench" (v1.1.4)[46] was used to benchmark the computational resources required per method with the *E. coli* dataset. Statistical tests were performed with Friedman rank sum test by R function "friedman.test" and Nemenyi post-hoc test[47] with R-package PMCMRplus[48]. Hardware specifications are provided in Table S3.

Benchmarking against Established Annotations

The accuracy of co-expression analysis methods were evaluated by comparing the proportion of gene pairs with genes matching to the same GO-IDs above a certain similarity-measure threshold. GO was chosen as ground truth data as it is commonly regarded as the gold-standard database in functional genomics that is regularly updated and reviewed[49], and is often used for benchmarking gene function inference methods[50]. Only GO-IDs for "Biological Process" GO terms were used. The GO-ID reference for *B. pseudomallei* K96243 was generated from genome assembly sequence annotations (NCBI RefSeq assembly GCF_000011545.1)[51]. The GO-ID reference for *E. coli* and *B. subtilis* were generated with data from QuickGO[52] after filtering for genes with "process" GO-terms. Gene pairs where at least one gene is not annotated to a GO-ID were removed.

The accuracy of co-expression methodologies was additionally evaluated by investigating whether or not gene pairs considered similar by each method share operons. This may additionally be informative in the use of co-expressed operons for BGC region refinement in COCOLOCUS. For the *E. coli* dataset, the above process was repeated to compare the proportion of gene pairs with genes annotated to the

same known operons above a certain similarity-measure threshold. Known Operon IDs (KOID) were obtained from the Known Operon Table from Operon Database Version 4 (ODB)[53,54].

Threshold selection

To evaluate the performance of each method in accurately identifying similar genes, the percentage of GO-ID identity and KOID identity were compared above a certain threshold between methods and against the background distribution. Since different similarity measures may be of different scales, a percentage of total gene pairs is selected across methods and datasets for consistency. Given that each expression dataset shares the same gene pairs across methods, cumulative identity intersects at a certain number of gene pairs across methods. For meaningful comparisons, the threshold must be higher than the intersection for all datasets. Furthermore, since GO-ID identity and KOID identity are strict evaluation metrics, a rigid correlation threshold was selected. Therefore, the threshold is selected so that similarity measures are at or below 0.9, due to the value being considered a robust and highly confident correlation cutoff[55,56]. With the above considerations, a threshold of the top 0.5% of gene pairs by similarity measure is selected (Figure S1).

Permutation test p-value

Permutation testing was performed to establish a statistical difference between the co-expression analysis methods and the background distribution. The background distribution was created with Monte Carlo sampling[57], or sampling *n* gene pairs from all filtered gene pairs without replacement. The sampled subsets contain a number of filtered gene pairs equal to 0.5% of the total number of filtered gene pairs per data set. 10,000 permutations were performed as recommended by literature[58].

Clustering and GO-Enrichment

GO-enrichment of clusters was performed to explore pathways identified by co-expression analysis methods and validate results to the experimental conditions[59]. Due to Bioconductor database availability, this was only performed for the *E. coli* dataset. Hierarchical clustering was performed with R function "hclust" and genes were clustered with function cutreeDynamic from R-package dynamicTreeCut (v1.63-1)[60]. A minimum cluster size of 30 genes was selected as a parameter as it is considered, in literature, the minimum number of genes for meaningful GO-enrichment[61].

The function compareCluster from Bioconductor package "clusterProfiler" (v4.19.3)[62] was used for GO-enrichment. Each gene cluster is enriched against the universe of all annotated genes in the expression matrix. The database org.EcK12.eg.db (v3.22.0)[63] was used as reference. Biological process (BP) was selected as the ontology parameter. The Benjamin Hochberd procedure (BH) was selected as the p-adjust method parameter.

To effectively compare GO-terms across clusters and methods, GO-terms were collapsed with the "simplify" function from clusterProfiler[62] with default parameters. Within each cluster, the top GO-terms by Gene Ratio were extracted. Tied top GO-terms of each cluster were all included.

Results and Discussion

Data acquisition and filtering

Data used to develop, test, and evaluate the explored workflows were selected from publicly available RNA-seq raw count datasets of at least 10 samples where both genes of at least 10% of gene pairs are GO-annotated. *B. pseudomallei* (K96243)[33], *E. coli* (K-12 MG1655)[38], and *B. subtilis* (str.168)[39] gene-expression profile datasets were used. The datasets contained 6335, 4305 genes, and 4636 genes for *B. pseudomallei*, *E. coli*, and *B. subtilis* datasets respectively. A brief overview of the experimental conditions of aforementioned datasets are reported in Table S1. Data processing statistics from low-expression filtering and zero imputations are provided in Table S2.

It may be important to note that published RNA-seq datasets appropriate for this analysis were challenging to curate, and analysis with plant and protist RNA-seq datasets were attempted but not included in this study due to computational and time limitations (Table S1). The connection of co-expression in bacteria to operon organization[64,65] , the gene regulation complexities in eukaryotes[66,67], and differences in experimental design[68,69] could introduce differences in best practices in co-expression analysis methodology between prokaryotes and eukaryotes. Thus, the scope of the results are only applicable to bacteria RNA-seq studies. Furthermore, the *E. coli* and *B. subtilis* datasets were not formally validated and quality control was limited to low-counts filtering as only raw-counts were available. Results should thus be interpreted with this in mind.

Comparative analysis of correlation distributions

Six total workflows were performed to compare correlation- and CoDA-based co-expression analysis normalization and transformation methods. Three correlation-based workflows available from COCOLOCUS and three CoDA-based workflows were used (Figure 1).

The similarity-measure distributions are inspected to examine normality assumptions relating to PCC. Density plots of similarities within subsets across the spectrum of gene expression (Figure 2) were created to preliminarily identify mean-correlation relationships or additional artifacts. It is worth noting here that proportionality-based measures do not assume normality.

Amongst all evaluated methods, distributions of similarities within subsets are the most skewed in Original and CLR across the similarity matrices. Mean residuals and PC1 residuals produce the most centered distributions across all matrices, though propr appears to produce more centered distributions around the median expression range (Middle 10% to self). Distributions tend to generally skew negative for within the bottom 10% and within the top 10% of genes. Correlations within the middle 10% of genes appear approximately centered while the correlations between the bottom 10% and top 10% of genes skew negative.

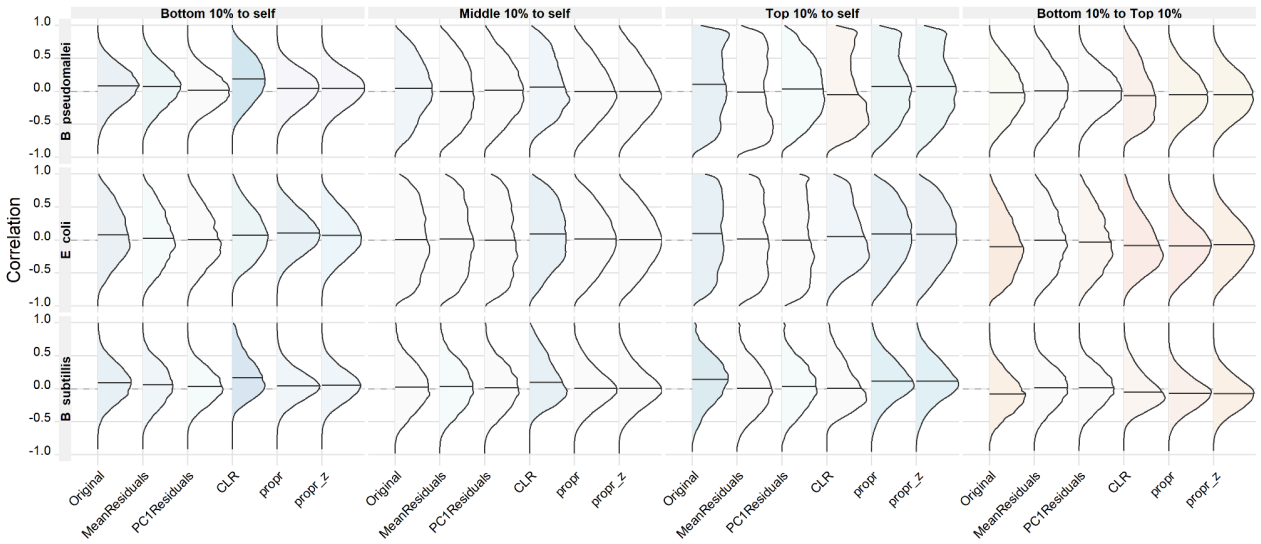

Figure 2. *Density plots of within-subset correlations across all datasets and methods. Genes are ordered by gene expression and similarities (correlation or proportionality) are calculated within the bottom 10%, middle 10%, and top % of genes, as well as between the bottom 10% and top 10% of genes by gene expression. Six density plots, one per method, are grouped by correlation within 10% subsets. Each row of density plots is of a different dataset. Density plots are shaded blue if skewing positive (median > 0) and red if skewing negative (median < 0). The median value of each bin is reported in Table S4.*

The distribution of correlations allows for a preliminary diagnosis of potential compositional biases and is especially informative in the context of methods using PCC, which does have a normality assumption. It is notable that the distributions are the most skewed in the Original and CLR methods, both of which use PCC and thus assume a normal distribution of correlations. This is especially notable amongst correlations of the top 10% of genes, which is consistent with findings regarding mean-correlation relationships in literature[45]. While correlation distributions offer an interesting insight into the data structure and allows for the diagnosis of potential compositional biases, it alone does not reflect the ability of each method to recover biological relationships between genes.

Proportionality-based methods more accurately identify similar genes

The accuracy of co-expression analysis methods was evaluated by matching genes to BP GO-terms and comparing the proportion of gene pairs with genes matching to the same GO-IDs. Data processing statistics for GO-ID filtering are reported in Table S2. The percentages of identical GO-IDs across the entire dataset are 2.8%, 4.9%, and 5.7% for *B. pseudomallei*, *E. coli*, and *B. subtilis* datasets respectively. Comparisons were made for the top 0.5% of gene pairs as well as across the collection of gene pairs. Delta percentage GO overlap values are reported as the difference between the percent GO overlap of subset and the percentages of identical GO-IDs across the dataset.

At the top 0.5% of gene pairs, both propr methods consistently outperform all other methods in identical GO-ID percentages across all datasets, but additional zero imputation does not appear to make a notable impact on propr results. CLR outperforms all conventional methods in two of three datasets, whereas Original, Mean Residuals, and PC1 Residuals perform relatively similarly to one another (Figure 3A). Based on permutation testing, all methods produce a significantly higher percentage of identical GO-ID than the background distribution across all datasets ($p < 0.0001$).

All three datasets show a similar pattern in percentage identical GO-IDs across the gene pairs, where all methods recover a high percentage of identical GO-IDs at gene pairs with high similarity measures, and the percentage drops rapidly. In all datasets, proportionality-based methods outperform methods using PCC for gene pairs with the highest similarity measures (Figure 3B-D). In addition, it is notable

that all methods other than "Original" identified anti-correlation in the *E. coli* dataset, with the propr methods recovering the highest percentage of identical GO-ID at the lowest similarity measures (strongest anti-correlation).

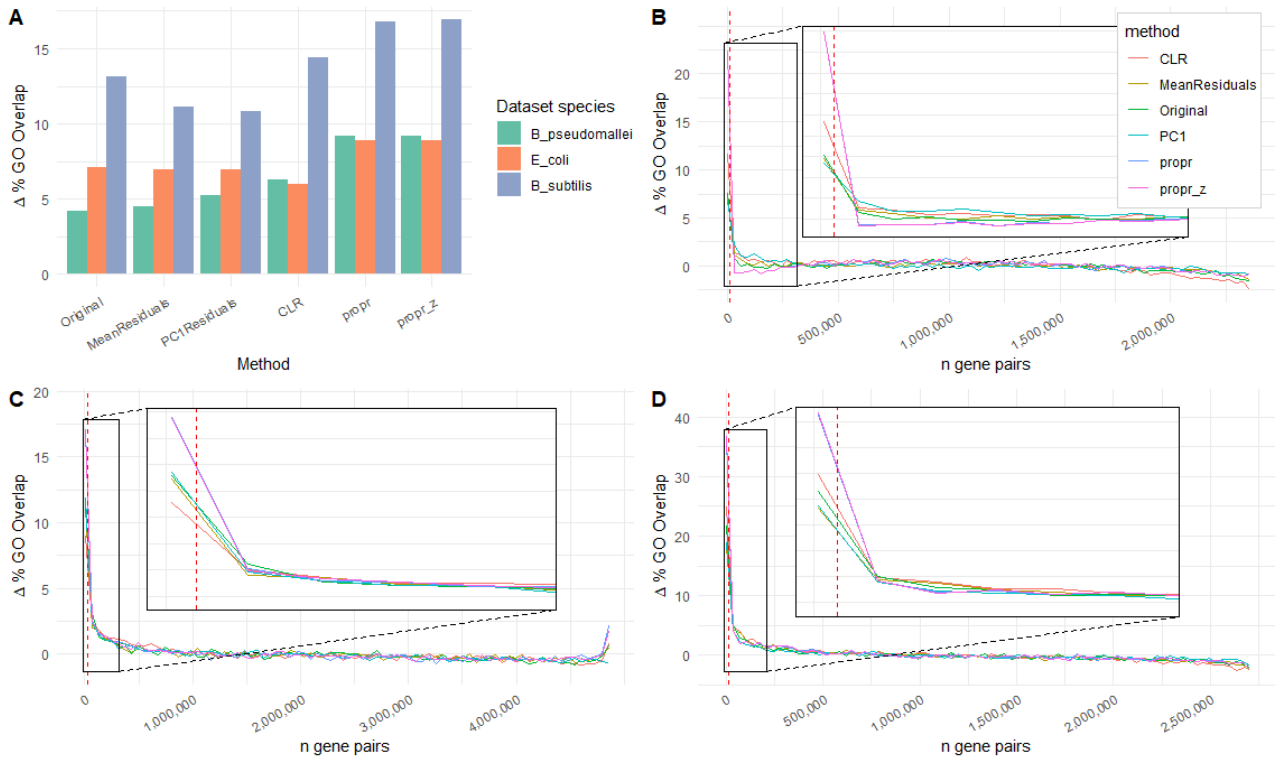

Figure 3. *Plots describing the accuracy of methods in identifying functional relationships across datasets, where "Δ % GO Overlap" refers to the difference between the percentages of identical GO-IDs in the reported subset and the mean percentage across the dataset. A: Bar plot of the Δ percentage GO Overlap amongst the top 0.5% highest correlating gene pairs per method. B-C: Line chart of the Δ percentage GO Overlap across all gene pairs, per method. Gene pairs are grouped 1000 bins in order of similarity measure. Lines are smoothed with Locally Estimated Scatterplot Smoothing (LOESS). A red dashed line is placed at n gene pairs = 0.5% of gene pairs for comparisons to Figure 3A. Figures 3B, C, and D are of B. pseudomallei, E. coli, and B. subtilis datasets, respectively.*

The accuracy of co-expression analysis methods in reflecting functional and structural relationships was additionally evaluated by matching genes to known operons and comparing the proportion of gene pairs with genes matching to the same known operon IDs (KOIDs). After filtering for gene pairs where both genes are annotated to at least one operon, 7074441 (92.2%) gene pairs were retained. Comparisons were made for the top 0.5% of gene pairs as well as across the collection of gene pairs. Delta percentage known operon overlap values are reported as the difference between the percent identical KOIDs of subset and the percentages of identical KOIDs across the dataset.

At the top 0.5% of gene pairs, both propr methods produce a higher percentage of shared operons between genes than all other methods (Figure 4A). All methods produce a significantly higher percentage of identical operons than the background distribution across all datasets ($p < 0.0001$). Similar to GO-IDs, all methods recover a high percentage of identical KOIDs at gene pairs with high similarity measures, and the percentage drops rapidly (Figure 4B). Here, proportionality-based methods outperform methods using PCC in gene pairs with the highest similarity measures.

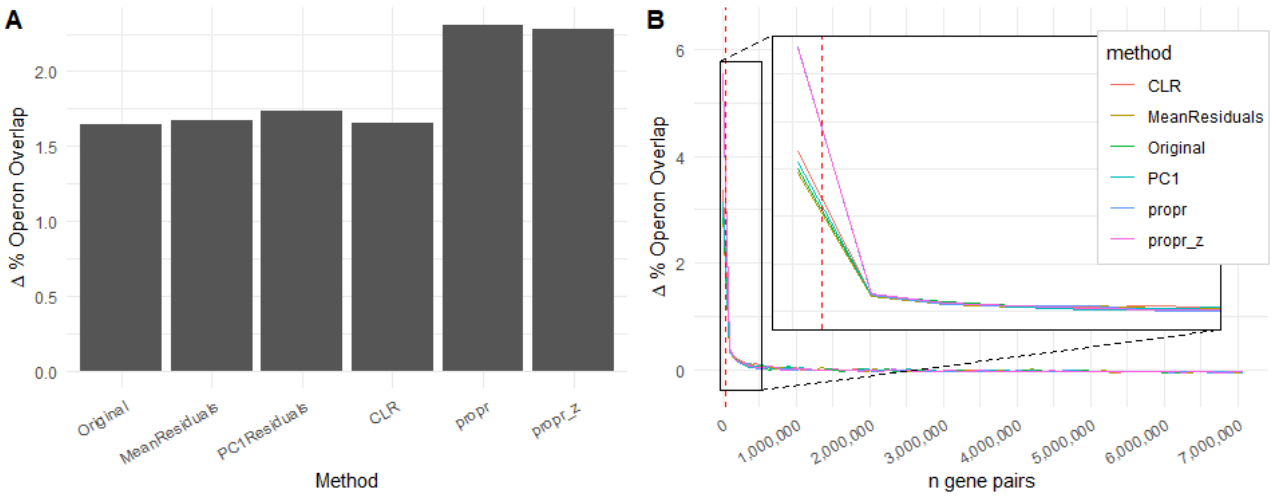

Figure 4. *Plots describing the accuracy of methods in identifying structural relationships with the E. coli dataset, where "Δ % Operon Overlap" refers to the difference between the percentages of identical KOIDs in the reported subset and the mean percentage across the dataset. A: Bar plot of Δ % Operon Overlap amongst the top 0.5% highest correlating gene pairs per method and the dataset means. B: Line chart of Δ % Operon Overlap across all gene pairs. Gene pairs are grouped 1000 bins in order of similarity measure. A red dashed line is placed at n gene pairs = 0.5% of gene pairs for comparisons to Figure 4A. Lines are smoothed with LOESS.*

Methods that produce accurate similarity measures are expected to return a high percentage of annotation overlaps at gene pairs of highest similarity, as genes that are co-expressed and correlated are likely to be involved in similar pathways[9] or be co-located[64,65,70]. Gene pairs with the lowest similarity are also expected to share GO-annotations, as anti-correlated genes may also be involved in similar pathways[71]. Based on these results, propr methods outperform correlation-based methods in their ability to accurately identify functional and structural relationships across datasets. This is consistent with work by Quinn et. al. and Lovell et. al. that the use of correlation-measures co-expression analysis do not most accurately identify similarities between genes[24,28].

The interpretability of these results, however, is limited by the completeness of annotations. Only the *E. coli* dataset retains more than half its gene pairs after filtering, which was also the only dataset where co-expression analysis detected anti-correlation. Sparse GO annotations have also been identified as a limitation to its use in benchmarking gene function inference in literature[50]. It may be therefore beneficial to further evaluate co-expression analysis methods against other databases such as the Kyoto Encyclopedia of Genes and Genomes (KEGG)[72] as additional sources of functional annotations. Furthermore, meaningful statistical comparisons of performance were not made between methods due to limited time, and between-method permutation testing at multiple similarity thresholds may be applied in future analysis for a more rigorous comparison.

Data transformation and normalization are computationally resource intensive

The runtime and memory benchmarking was performed to compare the computational resources required by each method using the *E. coli* dataset.

Statistical differences between methods are identified with Friedman rank sum test ($\chi^2(5) = 47.0$, $p < 0.001$). Nemenyi post-hoc test p-values are reported in Table S5. CLR has the shortest runtime whereas PC1 takes the longest to run, followed by propr + zCompositions, then propr. Methods "Original" and "Mean Residuals" have comparable runtimes. All conventional methods have comparable memory allocations, which are higher than all other methods. CLR has the lowest memory allocation and is overall the least computationally resource-intensive method.

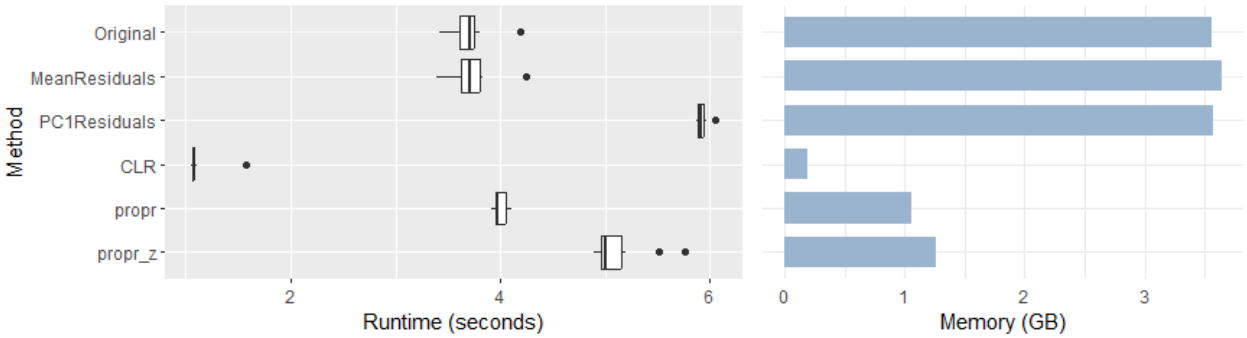

Figure 5. *Box plot and bar chart describing the runtime and memory allocation across 10 iterations of co-expression analysis with each method using the E. coli dataset.*

While the primary focus of this research concerns the methods' ability to identify biological information, computational resources may additionally inform method selection. The higher memory allocation by the methods Original, Mean Residuals, and PC1 Residuals can likely be attributed to its various layers of processing, specifically TMM normalization and spatial quantile normalization. This additionally affirm our initial concern regarding excessive data processing involved in correlation-based methods. Though differences in computational resources appear relatively small here, the pairwise computations would lead to demands scaling quadratically with genome sizes. To better compare the computational demands of each method, it may thus be interesting to further benchmark methods at different genome sizes and numbers of samples.

In addition, while zero imputation with zCompositions adds little to propr's ability to identify accurate functional and structural relationships, according to the boxplot, it is notably more computationally resource-intensive. This difference, however, was not identified in significance testing (Table S5). Regardless, this may further inform decisions between zero imputation methods.

Cluster analysis identify biological pathways consistent with experimental conditions

In order to compare methods by their ability to identify coherent gene clusters and to validate against experimental conditions, *E. coli* Genes were clustered based on the similarity matrices performed per each method, and GO-enrichment was performed for each cluster. After collapsing similar terms, top terms by Gene Ratio were extracted per cluster. For methods Original, Mean Residuals, PC1 Residuals, CLR, propr, and propr + zCompositions, 56, 57, 59, 61, 58, and 57 clusters were found, respectively.

Two GO-terms – "cytoplasmic translation" and "anaerobic respiration" – were shared amongst all methods. Extracted GO-terms primarily relate to protein synthesis, anaerobic respiration, aerobic respiration, carbon metabolism, amino acid metabolism, nucleotide biosynthetic processes, cell membrane, and ion transport (Table S6).

Cluster coherence was evaluated based on the gene-ratios of the clusters' top GO-terms. However, upon investigation, the assignment of GO-terms associated with genes after automated simplification appear arbitrary and inconsistent with established relationships between terms. For example, GO-terms relating to L-amino acid metabolic processes such as "L-amino acid biosynthetic process" and "L-amino acid catabolic process" were all presented in the same cluster despite terms being officially annotated as highly related terms[37,52] and thus potentially redundant. Therefore, it was difficult to meaningfully compare methods based on clusters and GO-term gene ratios.

On a preliminary analysis, representative GO-terms of clusters are consistent with genes relating to the experimental conditions. For example, anaerobic respiration was a term shared amongst clusters of all methods, which is consistent with the anaerobic experimental condition for the RNA-seq experiment. Ion transport genes, as another example, are consistent with the lithium acetate experimental condition[73]. However, as the priority of this study is in comparing the methods, an evaluation of the validity of the methods based on the GO-enrichment of the clusters was limited. A deeper dive into the GO-terms identified and their relevance to the experimental conditions, including a comparison of GO terms produced by clusters of each method, may act as an interesting avenue for further exploration.

General Discussion

Integrated results from the above evaluations shows that proportionality-based co-expression analysis methods by propr outperforms consistently correlation-based methods. Methods "Original" and "CLR" do not obey the normality assumption required for PCC, the former of which showing mean-correlation relationships[20] and the latter explained in literature where correlation-analysis is inappropriate for CoDA transformations due to constant-sum constraints[74]. Interestingly, despite correcting skewed distributions, using mean residuals and PC1 residuals do not appear to meaningfully or consistently improve the accuracy of the identification of similar genes compared to the "Original" method.

While propr methods outperform conventional correlation-based methods based on descriptive statistics, all methods recovered statistically significant functional- and structural-relationships. This was expected, as co-expression analysis remains frequently used for gene function predictions[65,75–77]. Applications of CoDA on discrete expression count data have been considered challenging, especially as it pertains to smaller expression datasets with a higher proportion of zero counts[31,32]. Thus, correlation-based methods may still be considered valid options for applications in smaller datasets. With that said, co-expression analysis has been shown to be more informative with larger datasets, and are frequently performed on multisource data[78,79]. It may thus be beneficial to further investigate the use of proportionality on multisource expression data for co-expression analysis across a larger number of samples as an alternative to correlation-based methods for more robust gene function predictions.

Conclusion

In this report, we demonstrate that within-subset correlation-distributions are often skewed in correlation-based methods, violating the normality assumption necessary for PCC calculations, begging further evaluation of the use of PCC in co-expression analysis. We show that proportionality-based similarity measures recover more accurate biological relationships between genes than correlation-based measures, which aligns with literature showing that RNA-seq data is compositional and should be analyzed with that assumption[24]. Of all evaluated methods, propr without additional zero imputation appears to be the most accurate co-expression analysis method. Based on the results of this study, compared to conventional co-expression analysis methods, proportionality-based co-expression analysis methodology does improve the detection of co-regulated pathways.

The study, however, is limited in its ability to claim statistical significance or its generalizability, due to the lack of significance testing between methods and its limitation to bacterial species and bulk RNAseq. While propr outperforms other evaluated methods based on descriptive statistics, further significance testing, such as additional permutation testing between methods or evaluations with various RNA-seq datasets of the same species may further solidify differences between methods. The methods are also only evaluated using bacterial datasets, and claims regarding the accuracy of proportionality-based methods cannot yet be generalized to other species.

In the context of improving the robustness of COCOLOCUS and other co-expression analysis tools, this report shows that conventional normalization algorithms and correlation-calculation techniques should be re-evaluated for co-expression analysis. Proportionality may be the answer to improving the robustness of functional-inferences of genes and gene clusters using RNA-seq data.

To further investigate the degree to which RNA-seq data as a whole is compositional, a simulated perturbation of RNA-seq data could be performed to examine changes in similarity-measure calculations. To further generalize the results of this study for practical bioinformatics applications, it should be repeated with scRNA-seq data, multisource expression data, and non-bacterial species and species with larger genomes, ideally with further optimized workflows.

Data Availability

All RNA-seq raw-counts used in this report can be found on NCBI GEO.
B. pseudomallei RNA-seq raw counts (accession number: GSE152295) may be accessed at https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE152295.
E. coli RNA-seq raw counts (accession number: GSE296075) may be accessed at https://www-ncbi-nlm-nih-gov/geo/query/acc.cgi?acc=GSE296075.
B. subtilis RNA-seq raw counts (accession number: GSE226559) may be accessed at https://www-ncbi-nlm-nih-gov/geo/query/acc.cgi?acc=GSE226559.

B. pseudomallei genome assembly and annotations (accession number: GCF_000959285.1) may be accessed at https://www.ncbi.nlm.nih.gov/datasets/genome/GCF_000959285.1/ . All data processing results as well as the code used for this project can be found at https://github.com/scarlettcsung/coexpression-thesis.

Code availability

All scripts used for the above methodology are available at https://github.com/scarlettcsung/coexpression-thesis/. This project used an unpublished version of COCOLOCUS included in the provided repository as “COBAIN”.

Acknowledgements

This project was made possible under the supervision of Robert Koetsier, Dr. Justin van der Hooft, and Professor Marnix Medema. This project was supported by the feedback of Master’s thesis students in the Wageningen University Bioinformatics group, particularly those in buddy groups Bogosort and Pandibles, as well as discussions with and feedback from Martijn Gobes.

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Supplementary Materials

Table S1. *Cell type (species, strain, tissue), replicates, conditions, data link, and reason for exclusion (if applicable) for all tested datasets. Notably, only bacterial datasets were used in this report. This was not a consequence of experimental design, but rather a result of dataset compatibility.*

Cell type	Replicates	Conditions	Data availability	Used? If not, why.
Burkholderia pseudomallei K96243	3	Control, pH, Iron, Hypoxia, Low nutrients, High NO, High NaCl, H2O2, Stationary, High Temp, Long Growth	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSM4610924	Yes
E. coli K-12 MG1655	2	Acetate, Anaerobic, LiAcet, M9 Glucose, Sorbital arginine, Paraquat, Stationary, TB	https://www-ncbi-nlm-nih-gov/geo/query/acc.cgi?acc=GSE296075	Yes
Saccharomyces cerevisiae S288c	3	Sodium cronate	https://www.ebi.ac.uk/gxa/experiments/E-MTAB-7471/Experiment%20Design	No, insufficient samples
Arabidopsis thaliana leaf	3	Salt, Heat, Both	https://www.ebi.ac.uk/gxa/experiments/E-GEOD-72806/Experiment%20Design	No, insufficient samples, genome too large to process
Bacillus subtilis str.168	3	Diamide, Ethanol, High Temperature, Low Temperature, Glucose to Malate Nutrient Shift, Malate to Glucose Nutrient Shift, Anaerobic Growth, Competence, Stationary Growth	https://www-ncbi-nlm-nih-gov/geo/query/acc.cgi?acc=GSE226559	Yes
Chlamydomonas reinhardtii	3	Nitrogen deprivation	https://www.ebi.ac.uk/gxa/experiments/E-GEOD-24365/Experiment%20Design	No, insufficient samples, genome too large to process

Table S2. *Data processing statistics of used datasets. filterLowExpression: number of genes and percentage of total genes removed for low expression. zCompositions imputation: number of counts zero-imputed by zCompositions. GO-ID filtering: number of gene pairs in dataset → number of genes remaining after GO-ID filtering, and percentage of gene pairs retained.*

Dataset	filterLowExpression	zCompositions imputation	GO-ID filtering
Burkholderia pseudomallei K96243	569 (8%) removed	2	16620495 → 2355535 (14.2% retained)
E. coli K-12 MG1655	388 (9% removed)	25	7669486 → 4853170 (63.3% retained)
Bacillus subtilis str.168	709 (15%) removed	9	7669486 → 4853170 (63.3% retained)

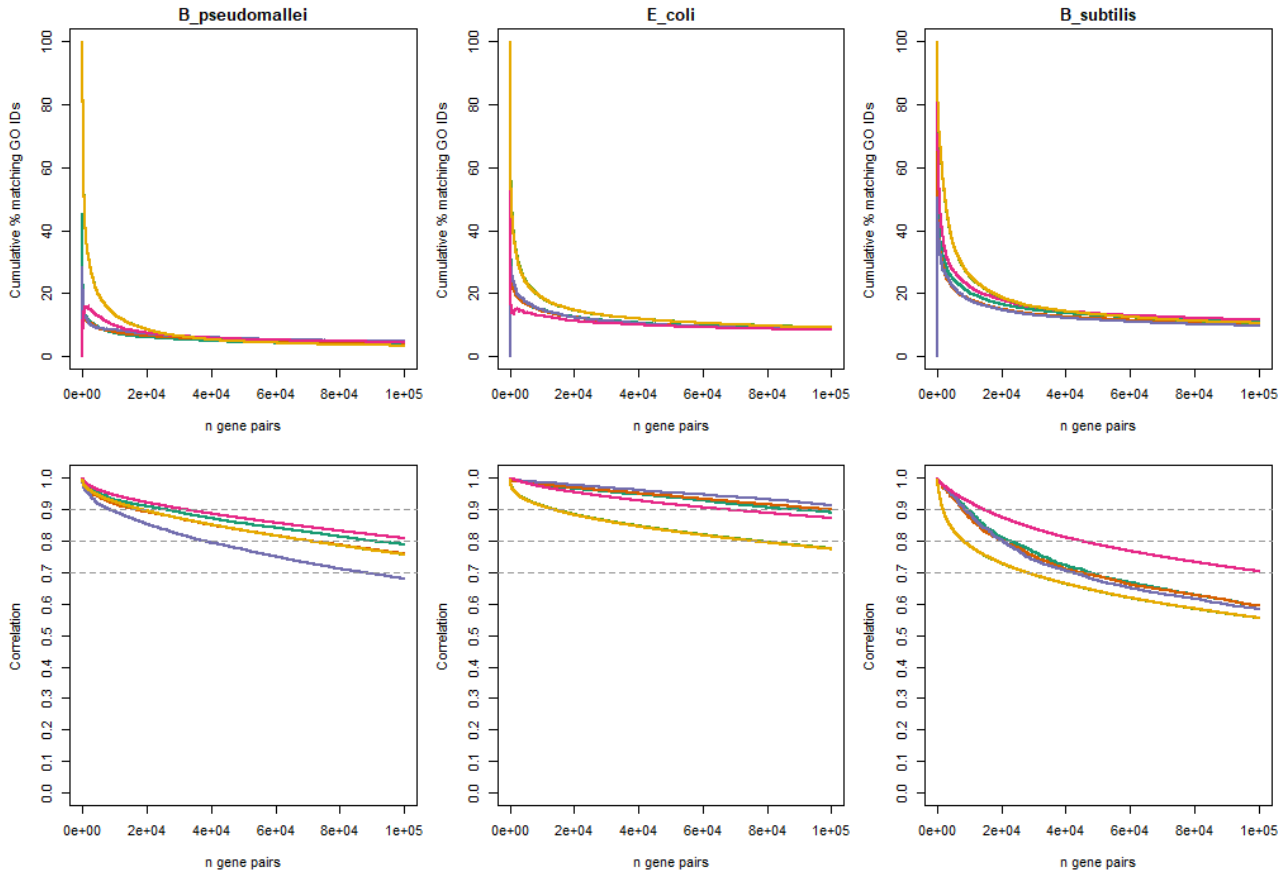

Figure S1. *Line charts referenced in threshold selection. Top row: Cumulative percent matching GO-IDs through the top 10000 gene pairs. Bottom row: Correlation across the top 10000 gene pairs. Primary considerations for the threshold is to be at a point where performance does not start overlapping (lines crossing over in top row charts). It should also be a threshold that can be consistently applied through all datasets. Since identical GO-IDs would imply strong co-expression, a high similarity measure is acceptable as a threshold. Therefore, annotation match comparisons are made at a similarity measure threshold of approximately 0.9.*

Table S3. *Hardware specifications in regards to computational benchmarking.*

Processor (CPU)	Intel(R) Core(TM) i5-10400 CPU @ 2.90GHz
Memory (RAM)	DDR4 3200MHz 16GB
Operating System	Windows 11 Pro 23H2

Table S4. *Median values of all within-subset correlation distributions.*

dataset	method	Bottom 10% to self	Middle 10% to self	Top 10% to self	Bottom 10% to Top 10%
B_pseudomallei	CLR	0.188664937	0.0607820292	-0.052937341	-0.065572709
B_pseudomallei	MeanResiduals	0.073287986	-0.0005981651	-0.015772697	0.00953977
B_pseudomallei	Original	0.082948836	0.04445371	0.104190466	-0.020376511
B_pseudomallei	PC1Residuals	0.01058329	0.0155296414	0.035250247	0.009540673
B_pseudomallei	propr	0.041898055	-0.0023559349	0.07297108	-0.055259538
B_pseudomallei	propr_z	0.038884323	-0.0022323272	0.07338728	-0.054251109
B_subtillis	CLR	0.164006993	0.0948472946	0.004634764	-0.050584804
B_subtillis	MeanResiduals	0.059884276	0.0292087228	0.003121928	0.012443373
B_subtillis	Original	0.090202154	0.0215011566	0.141347179	-0.079259952
B_subtillis	PC1Residuals	0.036253571	0.0175772689	0.031835411	0.015447692
B_subtillis	propr	0.045866302	0.0096318234	0.113040499	-0.069308446
B_subtillis	propr_z	0.052566624	0.0095907746	0.114763543	-0.074497079
E_coli	CLR	0.074414199	0.0921802317	0.047983853	-0.086427994
E_coli	MeanResiduals	0.02380385	0.0099021147	0.010752467	-0.004651624
E_coli	Original	0.077113574	0.003714569	0.096481577	-0.102765984
E_coli	PC1Residuals	0.009467844	-0.0025056735	-0.002347276	-0.036186803
E_coli	propr	0.10344296	0.009962804	0.088777672	-0.088482854
E_coli	propr_z	0.065821578	0.0071462832	0.084306047	-0.072778971

Table S5. *Matrix illustrating significance of runtime between all methods pairwise comparisons using Nemenyi-Wilcoxon-Wilcox all-pairs test for a two-way balanced complete block design.*

	CLR	MeanResiduals	Original	PC1Residuals	propr
MeanResiduals	3.94E-01	-	-	-	-
Original	3.94E-01	1.00E+00	-	-	-
PC1Residuals	3.40E-08	6.90E-04	6.90E-04	-	-
propr	1.06E-02	7.06E-01	7.06E-01	9.02E-02	-
propr_z	2.60E-05	4.74E-02	4.74E-02	8.39E-01	7.06E-01

Table S6. *Representative GO-terms per cluster across all methods. Methods are annotated with 1 if GO-term is present as a representative term of one of its clusters. The number of methods where the GO-term was present is reported under "method_sum".*

Description	Original	MeanResiduals	PC1Residuals	CLR	propr	propr_z	method_sum
cytoplasmic translation	1	1	1	1	1	1	6
anaerobic respiration	1	1	1	1	1	1	6
monocarboxylic acid catabolic process	1	1	1	1	0	1	5
non-proteinogenic amino acid metabolic process	0	1	1	1	1	1	5
biofilm formation	0	1	1	1	1	1	5
response to temperature stimulus	1	1	1	0	1	0	4
response to pH	1	1	0	0	1	1	4
L-amino acid metabolic process	1	1	1	1	0	0	4
proteinogenic amino acid metabolic process	1	1	1	1	0	0	4
aerobic respiration	0	1	1	0	1	1	4
lipopolysaccharide biosynthetic process	0	1	1	1	1	0	4
regulation of growth	0	1	0	1	1	1	4
anaerobic electron transport chain	0	0	1	1	1	1	4
single-species biofilm formation	1	0	0	1	1	0	3
membraneless organelle assembly	1	0	0	1	0	1	3
tRNA aminoacylation for protein translation	1	0	0	1	1	0	3
L-amino acid biosynthetic process	1	0	0	1	0	1	3
proteinogenic amino acid biosynthetic process	1	0	0	1	0	1	3
thiamine diphosphate biosynthetic process	0	1	0	1	1	0	3
bacterial-type flagellum-dependent cell motility	0	1	0	1	1	0	3
response to osmotic stress	0	1	1	0	1	0	3
de novo' IMP biosynthetic process	0	0	1	1	0	1	3
aerobic electron transport chain	1	0	0	0	0	1	2
DNA transposition	1	1	0	0	0	0	2
response to heat	1	1	0	0	0	0	2
putrescine catabolic process	1	0	0	0	1	0	2
monosaccharide transmembrane transport	1	1	0	0	0	0	2
transition metal ion transport	1	0	1	0	0	0	2
L-amino acid catabolic process	0	1	0	1	0	0	2
glucose catabolic process	0	1	0	1	0	0	2
response to copper ion	0	1	0	1	0	0	2
amino acid biosynthetic process	0	1	0	0	0	1	2
iron import into cell	0	1	0	0	1	0	2
methionine biosynthetic process	0	0	1	1	0	0	2
ribose phosphate biosynthetic process	0	0	1	0	0	1	2
siderophore metabolic process	0	0	1	0	0	1	2
iron ion transport	0	0	0	1	1	0	2
carbohydrate transport	0	0	0	0	1	1	2
lipid A biosynthetic process	0	0	0	0	1	1	2

nucleobase-containing small molecule interconversion	0	0	0	0	1	1	2
L-amino acid transport	0	0	0	0	1	1	2
energy derivation by oxidation of organic compounds	0	0	0	0	1	1	2
DNA-templated DNA replication	1	0	0	0	0	0	1
propionate metabolic process	1	0	0	0	0	0	1
aspartate family amino acid biosynthetic process	1	0	0	0	0	0	1
secondary metabolite biosynthetic process	1	0	0	0	0	0	1
lipopolysaccharide metabolic process	1	0	0	0	0	0	1
respiratory electron transport chain	1	0	0	0	0	0	1
SOS response	1	0	0	0	0	0	1
hexose metabolic process	1	0	0	0	0	0	1
nucleotide biosynthetic process	1	0	0	0	0	0	1
organic acid transmembrane transport	1	0	0	0	0	0	1
alpha-amino acid metabolic process	0	1	0	0	0	0	1
peptidoglycan metabolic process	0	1	0	0	0	0	1
L-arginine biosynthetic process	0	1	0	0	0	0	1
ribonucleoside triphosphate biosynthetic process	0	1	0	0	0	0	1
glycerol metabolic process	0	1	0	0	0	0	1
submerged biofilm formation	0	1	0	0	0	0	1
pyrimidine ribonucleotide biosynthetic process	0	1	0	0	0	0	1
phosphate ion transmembrane transport	0	1	0	0	0	0	1
quorum sensing	0	1	0	0	0	0	1
cell motility	0	0	1	0	0	0	1
cytochrome complex assembly	0	0	1	0	0	0	1
clearance of foreign intracellular nucleic acids	0	0	1	0	0	0	1
alpha-amino acid biosynthetic process	0	0	1	0	0	0	1
DNA metabolic process	0	0	1	0	0	0	1
aromatic amino acid family biosynthetic process	0	0	1	0	0	0	1
organic cation transport	0	0	1	0	0	0	1
phosphoenolpyruvate-dependent sugar phosphotransferase system	0	0	1	0	0	0	1
cell wall organization	0	0	0	1	0	0	1
organophosphate biosynthetic process	0	0	0	1	0	0	1
xenobiotic catabolic process	0	0	0	1	0	0	1
dipeptide transport	0	0	0	1	0	0	1
small molecule biosynthetic process	0	0	0	1	0	0	1
rRNA processing	0	0	0	1	0	0	1
iron-sulfur cluster assembly	0	0	0	1	0	0	1
translation	0	0	0	1	0	0	1
carboxylic acid transmembrane transport	0	0	0	1	0	0	1

ribosome biogenesis	0	0	0	1	0	0	1
monoatomic ion transmembrane transport	0	0	0	1	0	0	1
response to virus	0	0	0	1	0	0	1
ketone biosynthetic process	0	0	0	1	0	0	1
rRNA base methylation	0	0	0	0	1	0	1
monocarboxylic acid metabolic process	0	0	0	0	1	0	1
bacterial-type flagellum organization	0	0	0	0	1	0	1
cellular detoxification	0	0	0	0	1	0	1
hexose import across plasma membrane	0	0	0	0	1	0	1
DNA recombination	0	0	0	0	0	1	1
RNA methylation	0	0	0	0	0	1	1
alcohol biosynthetic process	0	0	0	0	0	1	1
purine ribonucleoside triphosphate biosynthetic process	0	0	0	0	0	1	1
lipopolysaccharide core region biosynthetic process	0	0	0	0	0	1	1

Appendix: Regarding Generative AI

Statement of Generative AI Use

Generative AI tools used in this project were ChatGPT (GPT-5, GPT-5.1, GPT-5.2 and GPT-5 Mini), Gemini 3, and DeepSeek-V3.2.

Part of the code used in this project was generated or edited by generative AI. Generative AI tools were used in debugging throughout the development of the workflows and evaluations. All scripts generated or edited by generative AI and their produced results were manually reviewed, checked against documentation of relevant R-packages and functions, and tested. Unit testing was also performed for select functions and included in the provided code repository. Inline documentation includes notes on where generative AI was used.

In report writing, generative AI tools were used to brainstorm phrasing, formulate report structure, and inspire titles. Prompts and responses used here are included at the end of this document.

Generally, generative AI tools were used for breaking down concepts and assisting in literature search. All information derived from text generated by LLM used in this project was validated against literature.

You Asked

How should I title my results portion sub-headings for my thesis report? my plan for the first section is to talk about the basic statistics of the comparisons, density plots (within subsets... like bottom 10% of genes vs itself etc) etc. but I'm not sure if htey should be one section, and how to title them

Gemini Fast

Titling a results section is a balancing act. You want to be descriptive enough that a reader can skim the headers and understand your "story," but professional enough to fit a thesis.

For your first section, combining basic statistics and density plots is a great move. They both serve the same purpose: **establishing the baseline characteristics of your data.**

Here are three different strategies for how to group and title these sub-headings:

Option 1: The "Functional" Approach (Standard)

This is the safest bet for a formal thesis. It tells the reader exactly what tools you used.

- **3.1 Descriptive Statistics and Global Distributions**
  - *Inside this:* You'd cover the mean/median comparisons and the overall density plots.
- **3.2 Comparative Analysis of Gene Subsets**
  - *Inside this:* You'd dive into the specific density plots for the bottom 10% vs. other cohorts.

Option 2: The "Narrative" Approach (Recommended)

This is often preferred by advisors because it highlights the *findings* rather than the *methods*. It makes the report much more readable.

- **3.1 Characterization of Global Expression Profiles**
- **3.2 Distributional Divergence Across Gene Expression Subsets**
  - *Note:* Using words like "Characterization" or "Divergence" tells the reader what the data *did*, not just that you made a plot.

Option 3: Combined (One Large Section)

If the basic stats and the density plots are short and lead to the same conclusion, keep them together to maintain momentum.

- **3.1 Statistical Distribution and Subset Comparison**

> **Tip:** If you go this route, use bolded lead-in sentences for your paragraphs (e.g., "**Global distribution trends.**" or "**Subset-specific density analysis.**") to help the reader navigate.

Should they be one section?

Yes, if the "basic statistics" are essentially just a setup for the density plots. If the statistics (like p-values or mean differences) are the evidence that justifies why you looked at the bottom 10%, keep them together.

No, if the basic statistics cover the whole genome and the density plots focus on a very specific biological phenomenon that requires its own lengthy interpretation.

Quick Naming Tips:

- **Avoid "Results of...":** It's redundant. We know they are results; you're in the Results chapter!
- **Be specific:** Instead of "Density Plots," try "Density Analysis of [Specific Variable]."
- **Keep it parallel:** If 3.1 starts with a noun (Characterization...), 3.2 should also start with a noun (Identification...).

Would you like me to help you draft a specific opening sentence for the "bottom 10%" section to help tie the statistics and plots together?

You Asked

how do you describe the subsets though?

Gemini Fast

Describing subsets in a heading requires a "Goldilocks" approach: you want to be more specific than "Subsets," but less wordy than a full sentence.

In a thesis, you should describe them based on the **criteria** used to create them (e.g., expression levels, rank, or biological categories).

Here are a few ways to phrase those sub-headings, depending on what those subsets actually represent:

1. The "Quantile" Approach

If your subsets are based on ranking (like your bottom 10%), use terms that describe their position in the distribution.

- **Analysis of Low-Abundance Gene Cohorts** (Refined way to say "bottom 10%")
- **Distributional Profiles Across Expression Deciles** (If you are comparing 10% blocks)
- **Comparative Dynamics of High vs. Low-Ranked Subsets**

2. The "Functional" Approach

If the subsets represent a specific state or characteristic.

- **Variation within Minimal-Expression Subsets**
- **Comparison of Extreme Percentile Distributions**
- **Divergence in Expression-Restricted Gene Groups**

3. The "Methodological" Approach

If you want to be very literal about how the data was sliced.

- **Density Analysis of Stratified Gene Populations**

- **Statistical Characterization of Tail-End Distributions**
- **Comparison of Expression Subgroups by Percentile Rank**

How to phrase it in the text (The "Lead-in")

When you move from the "Big Picture" stats to the "Subsets," you need a transition sentence. Here are three flavors:

- **The "Deep Dive" style:** *"To determine if global trends were preserved at the extremes of the dataset, genes were stratified into deciles for subset-specific density analysis."*
- **The "Focus" style:** *"Initial global comparisons necessitated a more granular look at low-expression genes, specifically the bottom 10th percentile..."*
- **The "Logic" style:** *"Recognizing that mean values can mask variance in low-frequency populations, we partitioned the data into [X] subsets..."*

A Pro-Tip on Terminology

Instead of saying "the bottom 10% vs itself," which sounds a bit confusing in a report, try:

- **"Internal consistency within the lowest decile"**
- **"Self-comparison of the 10th percentile cohort"**
- **"Intra-subset variance analysis"**

Would you like me to take a look at the specific variables you're comparing (e.g., Treatment vs. Control) so I can give you a more tailored heading?

You Asked

well ok so I'm doing co-expression analysis and thus I'm referring to correlations

Gemini Fast

That changes the framing significantly! In co-expression analysis, you aren't just looking at levels of expression; you're looking at the **strength and behavior of relationships.**

When you talk about the "bottom 10%," you are likely referring to the **weakest correlations** or perhaps genes with the **lowest connectivity** (k-diff).

Here is how you can title those sections using professional, co-expression-specific terminology:

---

1. Titling the "Basic Stats" Section

Since this is about correlation coefficients (like Pearson or Spearman), focus on the "Global" landscape.

- **3.1 Global Landscape of Co-expression Coefficients**
- **3.2 Distributional Properties of the Correlation Matrix**
- **3.3 Statistical Baseline of Gene-Pair Connectivity**

2. Titling the Subsets (The "Bottom 10%" etc.)

In co-expression, the "bottom" or "top" usually refers to **correlation strength** or **node degree**. You should use terms like "Strength," "Stringency," or "Strata."

- **3.2 Correlation Density across Coefficient Strata**
- **3.3 Comparative Analysis of Low-Strength Co-expression Subsets**
- **3.4 Stability of Weakly-Correlated Gene Pairs** (If your "bottom 10% vs itself" is a way of showing noise or baseline stability).
- **3.5 Distributional Divergence between High- and Low-Confidence Modules**

3. How to describe the "Bottom 10% vs. Itself"

In a thesis, saying "vs. itself" can be tricky. It usually implies you are looking at **Internal Consistency** or **Background Noise**.

Instead of: *"Density plots of bottom 10% vs. itself"* **Try:**

- **"Intra-subset Correlation Density within the Lowest Decile"**
- **"Homogeneity Analysis of Low-Correlation Gene Pairs"**
- **"Internal Distributional Profiles of Strength-Partitioned Subsets"**

Suggested Layout for your Results Section

Section	Suggested Title	What it covers
3.1	Global Distribution of Correlation Coefficients	Mean/Median correlations, the "big" density plot of all gene pairs.
3.2	Stratified Density Analysis of Co-expression Subsets	Breaking the correlations into tiers (Top 10%, Bottom 10%).
3.3	Internal Consistency of Low-Strength Gene Associations	The "bottom 10% vs itself" plots to show noise or baseline.

A Note on "Self-Comparison"

If your density plot of the "bottom 10% vs itself" is meant to show that these correlations are essentially random noise (centered around zero), you might want a heading that reflects that conclusion:

- *e.g.,* **"3.3 Characterization of Stochastic Noise in Low-Correlation Thresholds"**

Does "bottom 10%" in your case refer to the genes with the lowest average correlation, or are you picking the 10% weakest individual links in your network?

You Asked

I don't mean the bottom 10% but the idea of the subsetting itself. like bottom 10%, middle 10%, top 10% etc

Gemini Fast

Got it. Since you are performing this "slicing" across the entire range of your data (from the weakest to the strongest correlations), the key word you want to use is **Stratification** or **Binning**.

You aren't just looking at one group; you are looking at the **behavior of the data across different tiers of strength.**

Here are three ways to title these sections, depending on how much "weight" you want to give the subsetting:

Option 1: The "Analytical" Style (Focuses on the process)

Use this if the subsetting is a major part of your methodology to prove your results are robust.

- **3.1 Global Correlation Statistics and Distributional Trends**
- **3.2 Density Profiling Across Correlation Strata**
- **3.3 Comparative Analysis of Tiered Co-expression Subsets**

Option 2: The "Technical" Style (Focuses on the math)

This is very common in bioinformatics or computational theses.

- **3.1 Summary Statistics of the Co-expression Matrix**
- **3.2 Decile-based Stratification of Correlation Coefficients**
- **3.3 Distributional Consistency within Correlation Bins** (The "bins" are your 10% slices).

Option 3: The "Narrative" Style (Focuses on the insight)

Use this if the point of the subsetting is to show how the "behavior" of the genes changes as they become more/less correlated.

- **3.1 Statistical Characterization of Gene Connectivity**
- **3.2 Evaluation of Co-expression Density across Strength Gradients**
- **3.3 Intrinsic Variance within Correlation Percentile Groups**

How to describe the subsets in your prose

When you are writing the actual paragraphs, avoid saying "the 10% boxes." Instead, use these more formal terms to describe the subsets:

- **"Correlation Strata"** (Singular: Stratum)
- **"Coefficient Deciles"** (Specifically for 10% slices)
- **"Strength-partitioned cohorts"**
- **"Rank-ordered bins"**

A "Cheat Sheet" for your Sub-headings

If you want a clean, professional hierarchy, try this:

> **3.1 Global Distribution of Co-expression Coefficients** *(Cover the basic stats and the main density plot here)*
>
> **3.2 Comparative Density Profiling of Stratified Correlation Bins** *(Introduce the idea that you sliced the data into 10% increments to look at the "Top," "Middle," and "Bottom" behaviors)*

> **3.3 Intra-bin Homogeneity and Distributional Trends** *(Focus on the "subset vs itself" plots to show how stable the correlations are within those specific slices)*

Quick Tip: If you are using 10% slices specifically, the word "**Decile**" is your best friend. It's the mathematically precise term for 10% increments and makes your headings sound very rigorous.

Would you like me to help you write a "justification" sentence for why you chose to subset the data this way? (e.g., to ensure that your findings aren't just driven by the top-tier "outlier" correlations?)

You Asked

I see a lot of methods sections has an overview subsection. how would that apply here?

Gemini Fast

In a Results chapter, an **Overview** subsection acts as a "map" for the reader. While a Methods overview explains *how* you did the work, a **Results Overview** summarizes the *scope* of the data you generated before you dive into the nitty-gritty analysis.

It is particularly useful in a thesis to show the "Big Picture" (e.g., "We started with 20,000 genes, which resulted in X million correlation pairs").

How to apply an "Overview" to your Results

Think of this section as the "**Data Sanitization and Summary**" phase. You aren't interpreting deep patterns yet; you are describing the final dataset that came out of your pipeline.

Common titles for this section:

- **3.1 Overview of Co-expression Data Properties**
- **3.2 Summary of the Correlation Matrix and Data Quality**
- **3.1 Initial Characterization of Gene-Pair Associations**

What actually goes inside it?

- **Final Counts:** How many genes and samples survived your filtering?
- **Global Metrics:** What was the overall mean and median correlation? (This is where you put those "Basic Statistics" you mentioned).
- **High-Level Distribution:** A brief mention of the global density (the "all-vs-all" plot) to set the stage for the subsetting later.
- **The "Why":** A sentence explaining that because the data is [skewed/large/complex], you proceeded to analyze it via **stratification** (your 10% bins).

Integrating it with your Subsets

If you add an Overview, your structure becomes very logical and "flowy." Here is how it looks:

3.1 Overview of Global Co-expression Properties

Summarize the dataset size and the basic descriptive statistics (mean, variance, total pairs). This establishes the "baseline."

3.2 Stratified Analysis of Correlation Deciles

Explain that to understand the data's behavior at different levels of strength, you partitioned the correlations into 10% bins (deciles).

3.3 Distributional Density and Homogeneity within Strata

Show the density plots for the top, middle, and bottom 10%. This is where you compare the subsets to themselves and to each other.

When should you *not* include an Overview?

- **If your results are very short:** If you only have two pages of results, an overview might feel like filler.
- **If it repeats the Methods:** Don't explain *how* you calculated Pearson's r again. Just report the *values* that resulted from it.

A Wit-Infused Pro-Tip:

Think of the **Overview** as the "Trailer" for the movie. You're showing the best shots (the main stats) so the audience knows what kind of "film" (data) they are about to watch. The **Subsections** are the actual scenes where the plot thickens.

Would you like me to draft a short "Summary Table" template that you could put in your Overview section to display those basic statistics clearly?

You Asked

if I also want to present additional stats (filtering, zero imputation), would that go with the density plots or another subsection? I have an overview of RNA-seq data subsection. Is that good or should I name it something else/include other things in the overview section?

Gemini Fast

This is where "The Great Divide" between **Methods** and **Results** usually gets a bit blurry. The rule of thumb: **Methods** describes the *plan*, while **Results** describes the *outcome* of that plan.

Since filtering and zero imputation fundamentally change the "shape" of your data, you should definitely report the results of those steps.

---

1. The "Overview" Naming

"Overview of RNA-seq Data" is a bit broad. Since your thesis is focused on co-expression, you want the title to reflect that the data has been "cleaned" and is now ready for network analysis.

Better Title Options:

- **3.1 Data Pre-processing Outcomes and Quality Control**

- 3.1 Refinement of the RNA-seq Dataset for Co-expression Analysis
- 3.1 Characteristics of the Filtered and Imputed Expression Matrix

2. Where to put Filtering and Imputation?

These should definitely go in their own subsection (the Overview) **before** the density plots. You want to prove the data is "healthy" before you show how it correlates.

What to include in that Overview:

- **Filtering:** "Initial filtering removed X low-count genes, leaving Y genes for analysis."
- **Zero Imputation:** "Following imputation, the percentage of zero-values decreased from $A\%$ to $B\%$."
- **The Impact:** Briefly mention if imputation changed the global distribution (e.g., "Imputation restored the normality of the distribution, as shown in...").

3. Proposed Structure: The "Flow"

By separating the "data cleaning" from the "correlation behavior," your results will read like a logical story.

3.1 Data Pre-processing and Quality Assessment

- **Filtering results:** (How many genes/samples were lost).
- **Imputation effects:** (How the zeros were handled and the impact on data density).
- **Basic RNA-seq stats:** (Library sizes, count distributions).

3.2 Global Statistical Profile of Co-expression

- **Basic Correlation Stats:** (Mean/Median r values).
- **Global Density Plot:** (The distribution of all correlations).

3.3 Stratified Analysis of Correlation Deciles

- **The "Slicing":** Explain the 10% binning.
- **Subset Density Plots:** (Top vs. Middle vs. Bottom 10%).
- **Intra-subset Comparisons:** (The "vs. itself" logic).

Why this works:

1. **Transparency:** By putting filtering/imputation first, you show you aren't hiding "messy" data.
2. **Logic:** You can't have a correlation (3.2) without first having a filtered gene list (3.1).
3. **Clarity:** It keeps your density plot section focused on the **biological/mathematical relationships** rather than technical cleanup.

> **Note on Imputation:** If you have "Before vs. After" density plots for the imputation, those belong in section 3.1. The "Top/Bottom 10%" density plots for the *correlations* belong in 3.3.

Would you like me to help you phrase the specific "Results" of your imputation? (e.g., describing how it shifted the distribution or reduced noise?)

You Asked

I did also described the datasets themselves as they were obtained online

Gemini Fast

Since your data comes from online repositories, you are essentially dealing with three distinct phases: **Provenance** (where it's from), **Processing** (how you cleaned it), and **Profiling** (what it looks like now).

In a thesis, you want to move from the "External" (the public data) to the "Internal" (your specific findings) as quickly and logically as possible.

Here is how you can structure these "Results" sub-headings to keep everything organized:

---

1. The "Data Origin" Section

If the description of the online datasets is brief, you can roll it into your Overview. If you are comparing multiple datasets (e.g., GTEx vs. TCGA), give them their own header.

- **3.1 Acquisition and Characteristics of [Source, e.g., Public RNA-seq] Cohorts**
- **3.1 Overview of Meta-Analysis Datasets and Clinical Metadata**
- **3.1 Source Data Summary and Composition**

2. The "Cleanup" Section

This is where the filtering and imputation results live. You are showing the "Delta" (the change) between the raw online data and your analysis-ready data.

- **3.2 Impact of Filtering and Imputation on Transcriptome Sparsity**
- **3.2 Dataset Refinement: Quality Control and Missing Value Recovery * 3.2 Pre-processing Outcomes: From Raw Counts to Analysis-Ready Matrix**

3. The "Correlation" Sections

Now that the data is explained and cleaned, you move into the co-expression "story."

- **3.3 Global Statistical Profile of the Co-expression Landscape** (Basic stats/global density).
- **3.4 Distributional Dynamics Across Stratified Correlation Deciles** (The 10% bins).
- **3.5 Consistency and Variance within Strength-Partitioned Subsets** (The "vs itself" plots).

The "Master Plan" Structure

If you want to be extremely clear for your examiners, use this flow:

Section	Content
3.1	**Dataset Composition and Source Summary** (The online datasets, sample sizes, phenotypes).
3.2	**Data Refinement and Quality Assessment** (Filtering results and imputation of zero values).
3.3	**Global Descriptive Statistics of Gene Co-expression** (Mean/Median/Std Dev of correlations).
3.4	**Comparative Density Analysis of Correlation Strata** (The 10% subsetting and binning).

A Tip on "Online Datasets"

When describing datasets obtained online in your **Results** (as opposed to the Methods), focus on the **results of your inspection** of that data.

- *Example:* "Initial inspection of the [Name] dataset revealed a high degree of zero-inflation (X%), necessitating the imputation steps described in section 3.2."

What if I have too many sections?

If it feels like too many headers, you can combine 3.1 and 3.2 into one larger section:

- **3.1 Technical Characterization of the Study Population and Data Pre-processing**

Would you like me to help you write the transition sentence that links the "Imputation" results (3.2) to the "Correlation" results (3.3)? This is often the trickiest "bridge" to write!

You Asked

it's not a very long paper though... I worry that it's going to be too many categories/words

Gemini Fast

That is a very valid concern. In a shorter paper or thesis, "header bloat" can make the writing feel fragmented. You want a **streamlined** flow where the reader feels like they are moving through a single, cohesive argument rather than a checklist.

If you want to keep it tight, you can collapse these into **three main pillars**. Here is how to group them so it feels substantial but concise:

1. The "Data Foundation" (Everything before the math)

Instead of separate sections for the source, the filtering, and the imputation, group them into one "Input" section. This tells the reader: *"Here is what I started with and how I fixed it."*

- **3.1 Dataset Integration and Quality Refinement**
  - *Covers:* Online sources, filtering counts, and zero imputation.
  - *Why it works:* It frames the "online data" and "imputation" as a single preparation step.

2. The "Global Landscape" (The big picture)

This is where you transition from "genes" to "co-expression." You provide the high-level stats and the main density plot here.

- **3.2 Global Statistical Profile of Co-expression**
  - *Covers:* Mean/median correlations and the overall density distribution.
  - *Why it works:* It provides the "baseline" before you start slicing the data into pieces.

3. The "Granular Analysis" (The 10% bins)

This is your "deep dive." You don't need a separate header for every 10% slice. One header covers the whole strategy of subsetting.

- **3.3 Stratified Distribution across Correlation Deciles**
  - *Covers:* The subsetting logic, the density plots for top/middle/bottom, and the internal (vs. itself) comparisons.
  - *Why it works:* "Stratified" is a powerful word that justifies why you broke the data into 10% chunks.

The "Ultra-Lean" Alternative (2 Sections)

If the paper is truly short (e.g., under 15–20 pages total), you can even go down to **two** main results sections:

- **3.1 Data Acquisition and Pre-processing Outcomes**
  - (Source $\longrightarrow$ Filtering $\longrightarrow$ Imputation)
- **3.2 Distributional Analysis of Global and Stratified Co-expression**
  - (Basic Stats $\longrightarrow$ Global Density $\longrightarrow$ 10% Bins)

Pro-Tip for Scannability

Since you are worried about "too many words," use **bold lead-ins** at the start of your paragraphs instead of using sub-headers.

> **Impact of zero imputation.** We observed that imputing missing values shifted the global mean correlation from...
>
> **Correlation behavior in the lowest deciles.** When examining the bottom 10% of gene pairs, density plots revealed...

Summary Table for your 3-Section Plan:

Heading	Key Content
3.1 Dataset Refinement	Sources + Filtering + Imputation
3.2 Global Co-expression Profile	Basic Stats + All-vs-All Density
3.3 Stratified Decile Analysis	10% Subsets + Within-group Density

Does this three-pillar approach feel more manageable for the length of your report, or should we try to condense it even further?

You Asked

but zero imputation is part of the comparison i.e. I use propr and propr + zCompositions

Gemini Fast

Ah, that changes everything! Since **zCompositions** (imputation) and **propr** (the proportionality measure) are the actual variables you are testing against each other, they aren't "cleanup"—they are the **heart of your results.**

In this case, the imputation isn't a background step; it's a **treatment** you are applying to the data to see how it changes the outcome.

Here is the most efficient, "lean" way to structure this so the comparison is front and center:

1. The "Starting Point"

Keep the source data brief so you can get to the "action."

- **3.1 Dataset Composition and Pre-processing**
  - *Briefly mention:* Source data and basic filtering. This sets the stage.

2. The "Comparison" (The Meat)

Since you are comparing two specific methods (`propr` vs. `propr + zCompositions`), use a heading that highlights the **impact** of that choice.

- **3.2 Impact of Zero Imputation on Global Co-expression Distributions**
  - *What to put here:* This is where your basic stats and the "all-vs-all" density plots go. You are showing how the "raw" `propr` compares to the "imputed" version.

3. The "Deep Dive" (The Bins)

Use this section to show if the two methods behave differently at the extremes (the 10% bins).

- **3.3 Stability of Correlation Strata across Imputation Frameworks**
  - *What to put here:* Your 10% subsets. This is where you show if imputation "fixes" the bottom 10% or if it primarily affects the top-tier correlations.

Why this is better for a short paper:

- **It creates a "Versus" narrative:** By naming the methods (or the concept of imputation) in the header, the reader knows exactly what the "fight" is about.
- **It's lean:** Only three headers, but they cover the entire logical flow from *Source* $\longrightarrow$ *Global Change* $\longrightarrow$ *Specific Subset Change*.

The "Header + Table" Combo

To save words, you can put all your basic statistics (mean, median, etc.) for both `propr` and `propr + zCompositions` into a single table in section 3.2. This allows your text to focus on the **interpretation** of the density plots rather than listing numbers.

Suggested Phrasing for your "Bins" Section:

If you want to sound very "Pro," use this for the 10% section:

> "3.3 Comparative Analysis of Strength-Partitioned Deciles"

Inside that section, you can say:

> *"To assess how imputation influenced different tiers of association strength, we partitioned the coefficients into deciles. Density plots of the lowest decile (bottom 10%) revealed that..."*

Would you like me to help you write a "Results Summary" sentence that compares the two methods (with and without zCompositions) based on what you saw in those density plots?

You Asked

I'm comparing 6 methods, 2 of which involves additional zero imputation. I kind of want to include that with method descriptions and resulting statistics/density plots, but I'm not sure how to frame it.

Gemini Fast

Comparing six methods is a lot to juggle, especially when two have that "extra" imputation step. To keep a short paper from feeling like a list of six different things, you need to **group them by their relationship to each other.**

The most logical way to frame this is as a "**Core Method vs. Augmented Method**" comparison. You have your base methods, and then the "Imputed" variants.

Here is a lean, 3-section structure that handles all 6 methods without getting messy:

3.1 Dataset Acquisition and Preliminary Filtering

Keep this very short.

- **Content:** Where the data came from and how many genes you kept after filtering.
- **Why:** You need to establish the "input" before you talk about the "output" of the 6 methods.

3.2 Comparative Statistical Profiles of Co-expression Methods

This is where you introduce your 6 methods and their global behavior.

- **The Framing:** Instead of 6 separate paragraphs, group them. Talk about the "General Trends" across all methods, then highlight how the **imputation-based variants** (the 2 methods) differ from the **standard** ones (the 4 methods).
- **Visuals:** This is where your **Global Density Plots** and a **Summary Table** of mean/median correlations for all 6 methods live.
- **Suggested Title:** *Global Distribution and Density of Correlation Frameworks*

3.3 Sensitivity and Stability across Correlation Strata

This is where your 10% bins (deciles) go.

- **The Framing:** Use the "bins" to show which methods are "noisy" and which are "robust."
- **The "Bottom 10% vs. Itself" logic:** This is your "Stress Test." You are showing how each of the 6 methods handles the "weakest" links. Does imputation make the bottom 10% look more like real biological signal or just smoother noise?
- **Suggested Title:** *Performance Divergence within Stratified Correlation Deciles*

A "Short Paper" Formatting Trick: The Comparison Table

To save a massive amount of word count, use a table in Section 3.2 to house the "Basic Statistics."

Method	Imputation	Mean r	Median r	SD
Method 1	None	...	...	...
Method 2	zCompositions	...	...	...
...	...	...	...	...

How to frame the "Subsets" in the text:

In section 3.3, instead of describing 6 sets of plots, describe the **patterns**:

- *"While the top decile (top 10%) showed high concordance across all six methods, the bottom decile revealed significant divergence. Specifically, the methods incorporating **zCompositions** exhibited [more/less] dispersion compared to their non-imputed counterparts..."*

Summary of Titles (Lean & Professional):

1. **3.1 Source Data and Pre-processing Summary**
2. **3.2 Global Distribution of Co-expression Coefficients**
3. **3.3 Consistency of Association Strength across Stratified Deciles**

Would you like me to help you draft a "key findings" sentence that explains why the imputation methods look different from the other four in those density plots?

You Asked

can I just combine 3.2 and 3.3?

Gemini Fast

Yes, absolutely. In a shorter paper, combining them is actually the **smarter move**. It prevents you from repeating yourself and allows you to link the "Global" stats directly to the "Subset" behavior in one continuous argument.

By combining them, you frame the section as a **multi-scale analysis**: you're looking at the data from a bird's-eye view (global) and then zooming in (stratified).

Here are two ways to title and structure that "Mega-Section":

Option A: The "Comparative" Title (Focuses on the 6 methods)

- **3.2 Comparative Analysis of Co-expression Metric Distributions**

> How to write it:
>
> 1. **Global:** Start with the summary stats (Table) and the main density plot for all 6 methods.
> 2. **Transition:** Explain that because global averages can hide local variation, you stratified the data into deciles.
> 3. **Local:** Present the density plots for the 10% bins, specifically highlighting how the two **imputation-based** methods behaved differently from the other four.

Option B: The "Technical" Title (Focuses on the data structure)

- **3.2 Statistical Properties and Stratified Density of Correlation Frameworks**

> How to write it:
>
> 1. **The Overview:** Give the "raw numbers" for the methods.
> 2. **The Slicing:** Immediately move into the subsetting.

> 3. **The "Why":** Use the "bottom 10% vs itself" plots to prove which of the 6 methods are most stable/least noisy.

Why this is better for a short paper:

- **Eliminates Redundancy:** You only have to introduce the 6 methods once.
- **Better Flow:** You can say, *"While the global mean for [Method X] was 0.4, the stratified analysis revealed this was driven primarily by the top decile..."* (This links the two ideas in one sentence).
- **Cleaner Document:** Your Table of Contents looks tight and professional.

Example of the final "Short Paper" Structure:

- **3.1 Data Source and Refinement** (Online data + Filtering + Imputation steps).
- **3.2 Comparative Evaluation of Co-expression Distributions** (All 6 methods: Global stats + 10% Bins + Density plots).
- **3.3 [Next Section: e.g., Biological Validation or Gene Ontology]**

One Small Caveat:

Since 3.2 will now be quite "dense" (6 methods × multiple plots), make sure you use **clear figure captions** or **sub-labels** (e.g., **Figure 1A: Global Density, Figure 1B: Bottom Decile**) so the reader doesn't get lost in the data.

Would you like me to help you write the "Thematic Sentence" that opens this combined section? It's the sentence that tells the reader: *"We are looking at these 6 methods globally and then slicing them into 10% chunks to see who performs best."*

You Asked

how would i title the methods and results section reporting on results after comparing identical GO-IDs and operon-IDs across gene pairs?

Gemini Fast

Since you are comparing **GO-IDs** (Gene Ontology) and **operon-IDs** (specific to prokaryotic gene clusters), you are moving from pure math into **biological validation**. You are essentially asking: *"Do these high correlations actually mean the genes do the same thing or live on the same piece of DNA?"*

Here is how to title those sections clearly and concisely for a short paper:

---

1. Methods Section

In the Methods, you need to describe the "Ground Truth" or "Benchmark" you used to test your 6 methods.

- **2.X Functional and Genomic Validation Framework**
- **2.X Criteria for Functional (GO) and Structural (Operon) Mapping**
- **2.X Benchmarking Co-expression against Biological Annotations**

> **What to include:** Briefly explain how you defined a "match" (e.g., "A gene pair was considered functionally related if they shared at least one GO-ID...").

---

2. Results Section

In the Results, you are reporting the **accuracy** or **concordance** of your 6 methods against these biological markers.

Option A: The "Validation" Approach (Focuses on Truth)

- **3.3 Biological Validation via Functional and Structural Enrichment**
- **3.3 Concordance of Co-expression with GO-Terms and Operon Structures**

Option B: The "Comparative" Approach (Focuses on which method won)

- **3.3 Comparative Accuracy of Methods in Recovering Biological Relationships**
- **3.3 Evaluation of Co-expression Sensitivity to Genomic Context**

How to structure this within the section:

Since you have two different types of IDs (GO and Operon), you can use small sub-headings or bold lead-ins to keep them distinct without adding too many pages:

- **Functional conservation (GO-IDs):** Discuss how many high-correlation pairs shared GO terms. Which of the 6 methods had the highest "hit rate"?
- **Genomic proximity (Operons):** Discuss how many pairs were in the same operon. This is usually a "stronger" signal than GO-IDs—did the **imputation** methods help or hurt this?

Combining it into your "Lean" Paper Structure:

If you want to keep the whole paper to just 3 or 4 Results sections, it would look like this:

1. **3.1 Data Refinement and Pre-processing** (Sources, filtering, imputation).
2. **3.2 Distributional Analysis of Co-expression Frameworks** (The 6 methods, global stats, and the 10% bins).
3. **3.3 Functional and Genomic Validation of Gene Associations** (The GO-ID and Operon-ID results).

A Witty Note on "Identical IDs":

In your results, you might find that some methods are great at finding **Operons** (which are very tight and high-correlation) but bad at finding **GO terms** (which can be more diffuse). If that happens, a great title for that section would be:

> "3.3 Differential Sensitivity to Functional vs. Structural Gene Associations"

Would you like me to help you draft the sentence that explains why you used Operons specifically as a benchmark for your co-expression analysis? (Usually, it's because operons are the "gold standard" for co-expression in bacteria.)

You Asked

how much do i re-iterate what I've already said as introduction etc. in my discussion portion

Gemini Fast

In a thesis, the Discussion should feel like a **synthesis**, not a **sequel**.

The biggest mistake students make is "The Loop"—summarizing the Results, which summarized the Methods, which summarized the Intro. To avoid this, follow the "**10% Rule**": No more than 10% of any Discussion paragraph should be a recap of the data. The rest should be interpretation.

Here is how to handle the re-iteration without being repetitive:

---

1. The "Intro" Recap: Re-frame, don't Repeat

In the **Introduction**, you stated a problem (e.g., "Zero-inflation biases co-expression"). In the **Discussion**, you state the *solution* your data provided.

- **Avoid:** "As stated in the introduction, RNA-seq data has many zeros." (We know.)
- **Instead:** "Our results demonstrate that the zero-inflation issues identified earlier are most effectively mitigated by [Method X] rather than [Method Y]."

2. The "Results" Recap: The "Evidence" Technique

Never list your results again. Instead, use them as **evidence** for a larger claim.

- **Avoid:** "Method A had a mean correlation of 0.65 while Method B had 0.4."
- **Instead:** "The significantly higher mean correlation observed in Method A suggests a higher sensitivity to [biological factor], though the stratified analysis in the bottom decile indicates this may include stochastic noise."

---

3. A "Lean" Discussion Structure

Since your paper is short, keep your Discussion focused on **three specific layers**:

Layer 1: The "What happened?" (Brief)

Open with a very short paragraph (3–4 sentences) summarizing the main takeaway.

- *Example:* "This study compared six co-expression frameworks to determine the impact of zero-imputation on bacterial gene networks. We found that while imputation smooths global distributions, its primary value lies in stabilizing the lowest correlation deciles."

Layer 2: The "Why does it matter?" (The Meat)

This is where you compare your 6 methods to the **GO-IDs** and **Operon-IDs**.

- Link your math to biology. If the imputation methods found more Operon-IDs, *why*?
- This is where you bring in **outside literature**. "Our finding that [Method X] recovers more operons aligns with Smith et al. (2024), who found that..."

Layer 3: The "So what?" (The Conclusion)

End with the limitations and the "Future Work."

- *Example:* "While our analysis focused on [Dataset X], the consistency of the decile-stratification approach suggests it could be a standard QC step for any co-expression pipeline."

---

4. Use "Mirroring" to stay organized

To keep the paper cohesive, try to mirror the order of your Results in your Discussion.

If Results Section 3.2 was...	Then Discussion Section 4.1 should be...
Statistical Distributions	Methodological Divergence & Noise
GO/Operon Validation	Biological Relevance & Accuracy

Summary Checklist:

- **Did I mention a p-value or a specific mean?** If yes, move it back to Results or delete it. Use "significantly higher" or "marginal increase" instead.
- **Am I citing new papers?** A good Discussion should introduce new citations that help explain *your* results.
- **Am I answering the "Why"?** If you are just saying "what" happened, you're still writing the Results.

Would you like me to help you write the "opening hook" for your Discussion that connects your 6-method comparison back to the "big picture" of your thesis?

You Asked

but would I say things like "we aimed to answer ... because ..." or something like that?

Gemini Fast

In a short paper, you want to avoid sounding like you are back at the starting line. Using "We aimed to answer..." can feel a bit like you're re-reading the syllabus to the professor.

Instead of stating what you **aimed** to do, state what you **achieved** or **determined**. This shifts the focus from your *intentions* (Intro) to your *contributions* (Discussion).

Here is how to transform "Intro-speak" into "Discussion-speak":

The "Status Upgrade" Strategy

Stage	The "Intro" Version (Aims)	The "Discussion" Version (Conclusions)
Objective	"This study aimed to determine if zero imputation improves co-expression accuracy..."	"Our comparison demonstrates that zero imputation primarily enhances the recovery of structural gene clusters (operons) by..."
Motivation	"Because RNA-seq data is sparse, we wanted to see..."	"The sparsity inherent in the source data necessitated the stratified approach, which revealed that..."
Methods	"We chose six methods to evaluate..."	"The divergence observed among the six methods highlights the sensitivity of bacterial networks to..."

Three Ways to Open the Discussion

Since you mentioned you are worried about word count and repetition, choose the "Opening" that best fits your narrative:

1. The "Problem Resolved" Opening (Strong & Direct)

> "While the impact of zero-inflation on RNA-seq analysis is well-documented, its specific effect on the stability of co-expression deciles has remained unclear. This study resolved this by comparing six metrics, finding that..."

2. The "Biological Discovery" Opening (Focuses on the GO/Operons)

> "The alignment of high-strength correlation pairs with known operon structures confirms that [Method X] provides the most biologically relevant representation of the bacterial transcriptome."

3. The "Methodological Verdict" Opening (Focuses on the 6 methods)

> "A comparison of six co-expression frameworks revealed that the inclusion of `zCompositions` significantly reduces stochastic noise in low-strength associations, as evidenced by the density shifts in the bottom 10% of gene pairs."

When is "We aimed to..." actually okay?

You can use it **once** at the very beginning of the Discussion as a "bridge," but then immediately pivot to the results.

> **Example:** "We aimed to identify the most robust co-expression metric for sparse bacterial data. **Our findings indicate** that the choice of metric is less critical for high-abundance genes than it is for the lower deciles, where..."

A Pro-Tip on "Because"

Instead of "We did X because Y," try using "**consistent with**" or "**suggests.**"

- **Weak:** "We looked at the bottom 10% because it is noisy."
- **Strong:** "The instability observed in the bottom 10% of correlations **suggests** that global metrics alone are insufficient for quality control in bacterial networks."

Would you like me to help you draft the final "Conclusion" sentence of your paper? It usually summarizes the most important method among the 6 you tested.

You Asked

do supplemental materials go before or after the references?

Gemini Fast

In most academic contexts, **Supplemental Materials go after the References.** However, the "where" depends slightly on whether you are submitting this to a journal, a university department, or a digital repository. Here is the standard breakdown:

1. The Standard Thesis/Paper Order

1. **Main Body** (Intro, Methods, Results, Discussion, Conclusion)
2. **References** (Starts on a new page)
3. **Appendices / Supplemental Materials** (Starts on a new page after the references)

2. Exceptions to the Rule

- **Journal Submission:** Many journals require you to upload Supplemental Materials as a **completely separate file** (Excel sheets, high-res images). If it is included in the main PDF, it almost always follows the references.
- **University Guidelines:** Some specific thesis templates place "Appendices" before the references so that the bibliography is the very last thing the reader sees. **Check your university's "Formatting Guide" PDF**—it usually has a "Order of Elements" page.

Pro-Tips for Supplemental Materials

- **Numbering:** Use a different numbering system to distinguish them from your main figures.
  - *Main Body:* Figure 1, Figure 2
  - *Supplemental:* **Figure S1, Figure S2, Table S1**
- **The "Stand Alone" Rule:** Each supplemental figure or table should have a caption that is descriptive enough to be understood without reading the main paper.
- **Internal Citations:** In your Results section, make sure you point to them!
  > *"Global stats were consistent across all trials (see **Supplementary Table S1**)."*

How to format the heading:

When you start the supplemental section, use a clear, centered title page or a bold heading:

Supplementary Material

Title of Thesis: [Your Title] **Author:** [Your Name]

Since your paper is short, are you putting your "Bottom 10% vs itself" density plots in the Supplemental section, or are they important enough to stay in the main Results? (If they show the "noise" you're trying to fix, they usually stay in the main body!)

You Asked

I can't think of a good title for this project...

Gemini Fast

Since your project involves a 6-way methodological "battle," zero-imputation, and biological validation via operons/GO-terms, your title needs to signal both **technical rigor** (the math) and **biological utility** (the genes).

Here are a few title options categorized by the "vibe" of your report:

1. The "Benchmarking" Title (Standard & Academic)

This is the most professional approach. It tells the reader exactly what you compared.

- **A Comparative Evaluation of Co-expression Frameworks for Sparse Bacterial Transcriptomes**
- **Benchmarking Correlation Metrics and Zero-Imputation Strategies in RNA-seq Co-expression Networks**
- **Performance Analysis of Six Co-expression Methods for Functional Gene Association**

2. The "Finding-Focused" Title (Punchy & Direct)

Use this if you want the title to reflect your main conclusion (e.g., that imputation matters).

- **Impact of Zero-Imputation on the Stability and Biological Accuracy of Bacterial Co-expression Networks**
- **From Noise to Signal: Assessing Stratified Correlation Deciles in Imputed RNA-seq Data**
- **Evaluating Genomic and Functional Concordance in Strength-Partitioned Co-expression Matrices**

3. The "Methodological" Title (Short & Clean)

If your paper is short and technical, keep the title lean.

- **Stratified Analysis of Co-expression Stability in Bacterial RNA-seq Data**
- **Assessing the Recovery of Operon Structures across Six Co-expression Metrics**
- **Stability and Validation of Imputed Co-expression Networks**

How to Build Your Own Title (The "Modular" Way)

If none of those feel quite right, mix and match one phrase from each column:

Column A (The Action)	Column B (The Subject)	Column C (The Context)
Comparative Analysis of	Zero-Imputation Strategies	in Bacterial Transcriptomics
Evaluating the Stability of	Co-expression Deciles	for Functional Mapping
Benchmarking Six	Correlation Frameworks	for Operon Recovery

Example result: *Comparative Analysis of Correlation Frameworks for Operon Recovery in Bacterial Transcriptomics.*

A Tip for Your Title Choice:

If you found that **operons** were the most important "truth" you recovered, make sure the word "**Operon**" or "**Genomic Structure**" is in the title. It makes your project sound much more specific and valuable than just "looking at data."

Would you like me to help you refine one of these once you decide which "finding" was the most important part of your results?

You Asked

I want an emphasis on compositionality and correlation vs proportionality

Gemini Fast

This adds a sophisticated layer to your title. Because you are using `propr` and `zCompositions`, you are specifically addressing the **compositional nature** of RNA-seq data (where counts are relative, not absolute).

In the world of compositional data analysis (CoDA), the "Proportionality vs. Correlation" debate is a major selling point. Here are several options that highlight that specific tension:

1. The "Formal Academic" Approach

These are best if your thesis is being submitted to a bioinformatics or statistics-heavy department.

- **Beyond Correlation: Evaluating Compositional Proportionality in Bacterial Co-expression Networks**
- **A Comparative Study of Proportionality and Correlation Metrics for Sparse Transcriptomic Data**
- **Assessing the Impact of Zero-Imputation on Compositional Co-expression Frameworks**

2. The "Finding-Focused" Approach

Use these if you found that proportionality (like ρ or ϕ from `propr`) performed better than standard Pearson/Spearman correlation.

- **Proportionality vs. Correlation: Recovering Biological Signal in Compositional RNA-seq Datasets**
- **Superiority of Proportionality Metrics in the Recovery of Bacterial Operon Structures**
- **Mitigating Compositional Bias: A Benchmarking of Six Co-expression Frameworks**

3. The "Methodological Deep-Dive"

These highlight the "stratified/binning" work you did.

- **Stratified Stability Analysis of Proportionality and Correlation in Imputed Bacterial Networks**
- **Evaluating the Sensitivity of Compositional Metrics across Correlation Deciles**

Why "Proportionality" and "Compositionality" are Power Words:

1. **Compositionality:** Signals that you understand that RNA-seq data is "closed" (it sums to a constant/library size) and that standard correlation can be mathematically invalid.
2. **Proportionality:** This is the specific solution to the compositionality problem. Using this word tells the reader you are using modern, robust alternatives to Pearson/Spearman.

My Recommendation for a "Best All-Rounder":

> "Evaluating Proportionality and Correlation: A Comparative Study of Compositional Co-expression in Bacterial Networks"

Why this works:

- It sets up the **battle** (Proportionality vs. Correlation).
- It mentions the **framework** (Compositional).
- It mentions the **subject** (Bacterial Networks).
- It's concise but sounds very "high-level."

One Quick Question for Your Title:

In your results, did the **proportionality** metrics (like `propr`) actually perform better than the standard **correlation** metrics when it came to finding the GO-IDs and Operons? If so, we can make the title even more assertive, like: *"The Efficacy of Proportionality Metrics for..."*

Would you like me to help you pick the best "Power Word" to describe your findings in the abstract based on this title?

You Asked

Proportionality as a promising method to address gene-expression data compositionality for more robust functional inference by co-expression analysis

I want to convey this idea but it's quite wordy for a thesis title, no?

Gemini Fast

It is a bit wordy, but the **logic** is perfect. You've captured the "Problem → Solution → Outcome" flow beautifully.

In a thesis title, you can trim the "filler" words (like "as a promising method to address") and replace them with more active, academic phrasing. You want to turn your **sentence** into a **statement**.

Here are four ways to "shrink" your idea while keeping the exact same meaning:

1. The "Standard Academic" (Balanced)

> Proportionality vs. Correlation: Enhancing Functional Inference in Compositional RNA-seq Networks

- **Why it works:** It sets up the comparison immediately and identifies the ultimate goal (Functional Inference).

2. The "Direct & Assertive" (Shortest)

> Robust Co-expression Analysis via Proportionality: Mitigating Compositionality in Bacterial Transcriptomes

- **Why it works:** It moves the "Solution" (Proportionality) to the front and uses "Mitigating" as a strong, active verb.

3. The "Methodological" (Very Professional)

> Evaluating Proportionality for Robust Functional Inference in Compositional Co-expression Networks

- **Why it works:** It sounds like a formal investigation. It keeps your "Functional Inference" goal but makes it more concise.

4. The "Colon" Style (Classic Thesis Format)

> Addressing Compositionality in Co-expression: Proportionality as a Framework for Robust Functional Inference

- **Why it works:** It separates the **Problem** (Compositionality) from the **Solution** (Proportionality).

Key Edits Made:

- "**As a promising method to address**" → became "**Mitigating**" or "**via**".
- "**Gene-expression data compositionality**" → became "**Compositional RNA-seq**" or "**Compositionality**".
- "**For more robust**" → became "**Enhancing**" or simply "**Robust**".

Which one should you choose?

- If your results **strongly prove** proportionality is better: Go with **Option 2**.
- If your thesis is a **balanced comparison** of the 6 methods: Go with **Option 1**.

Pro-Tip: If you feel the title is still too long, you can often move the "Bacterial" part to the first sentence of your abstract. However, keeping it in the title is usually better for searchability.

Would you like me to help you refine the "Abstract" opening based on whichever title you like best?

You Asked

"useful genes" seem like a bad phrasing.

Gemini Fast

You are absolutely right. "Useful genes" sounds a bit like they are tools in a shed rather than biological entities. It lacks the scientific precision your thesis deserves.

In the context of co-expression and functional inference, you are usually referring to genes that are **informative**, **biologically relevant**, or **well-characterized**.

Here are several ways to replace "useful genes" depending on what you actually mean by "useful":

1. If "useful" means they have a known function (GO-IDs)

- **Annotated genes** (The most common term in bioinformatics).
- **Functionally characterized genes**.
- **Known biomarkers** (If they relate to a specific condition).

2. If "useful" means they show strong, non-noisy signals

- **Informative features**.
- **High-variance transcripts**.
- **Robustly expressed genes**.
- **Differentially associated genes**.

3. If "useful" means they are part of a specific structure (Operons)

- **Genomically clustered genes**.
- **Syntenic gene pairs**.
- **Co-regulated loci**.

How to swap it in your title or text:

Instead of:

> "...for the identification of **useful genes** in co-expression networks."

Try:

> - "...for the identification of **functionally relevant gene associations**."
> - "...enhancing the recovery of **biologically informative gene modules**."
> - "...improving the detection of **functionally annotated gene pairs**."

Refining your Title (The Final Polish)

Taking your previous idea and swapping out the "useful" concept, here is a punchy, professional version:

> "Proportionality for Robust Functional Inference: Addressing Compositionality in Bacterial Co-expression Networks"

Why this is better:

- "**Functional Inference**" replaces the need to say "finding useful genes." It implies that your goal is to infer what the genes do (their function) based on the network.
- It sounds like a definitive piece of research rather than a general observation.

A Wit-Infused "Peer" Observation:

In biology, every gene thinks it's "useful"—even the "junk" DNA is having a PR crisis lately. Using "**Functionally Relevant**" is the ultimate academic "polite" way of saying some genes are more interesting to your study than others.

Would you like me to help you check a specific sentence where you used "useful genes" so we can find the exact right scientific replacement for that context?

You Asked

it's in the context of candidate gene identification via functional predictions

Gemini Fast

In that context, "useful genes" is definitely too vague. You are looking for genes that haven't been fully explored yet but show high potential for specific roles based on your model.

Here are the best professional alternatives for "**candidate gene identification via functional predictions**":

1. The "Standard" Terms

These are the bread and butter of bioinformatics papers:

- **Putative functional candidates:** "Putative" is the perfect academic word for "we think it does this, but it hasn't been proven in a wet lab yet."
- **Novel gene candidates:** Use this if you are finding genes that weren't previously associated with a specific pathway.
- **High-priority candidates:** This implies your method has "ranked" them as the most likely to be biologically interesting.

2. The "Discovery-Focused" Terms

If your goal is to bridge the gap between "unknown" and "known":

- **In silico predicted candidates:** Highlights that the identification was done via computational methods.
- **Uncharacterized transcripts of interest:** Specifically refers to genes with no current GO-IDs that your method suggests are important.
- **Candidate loci for functional validation:** This is a great "closer" for a paper—it tells the reader, "These are the genes someone should go test in a lab."

3. The "Network-Specific" Terms

Since you are doing co-expression/proportionality:

- **Network-inferred candidates:** Highlights that the "usefulness" comes from their position in the network.
- **Functionally synonymous gene pairs:** If you are focusing on the fact that they behave identically in your proportionality analysis.

How to use them in a sentence:

Instead of:

> "Our method helps find **useful genes** through functional predictions."

Try:

> - "The proposed framework facilitates the **prioritization of novel gene candidates** for subsequent functional validation."
> - "We demonstrate that proportionality analysis improves the **identification of putative functional candidates** within sparse datasets."
> - "This approach enhances the **recovery of biologically informative transcripts** from uncharacterized co-expression modules."

Which one fits your "Title" or "Conclusion" best?

- If you are focusing on the **accuracy** of your predictions: "**Putative functional candidates.**"
- If you are focusing on **saving time** for future researchers: "**Prioritizing gene candidates.**"
- If you are focusing on **compositionality** fixing the data: "**Informative gene associations.**"

Would you like me to help you draft a "Future Work" sentence using these terms to show how your candidate genes should be tested next?

You Asked

For my introduction, I am trying to transition from a discussion about what co-expression analysis is and how it is performed towards a bias found in co-expression analysis found while developing a software tool. I am not really sure how to make that transition, can you help?

ChatGPT gpt-5-mini

Absolutely! You can make the transition smoothly by first acknowledging the strengths and widespread use of co-expression analysis, then introducing the nuance that not all results are free from bias. Essentially, you're moving from "what it is and how it works" to "what issues can arise in practice." Here's a structure you could follow in a paragraph:

---

1. **Start with a brief recap:** One or two sentences summarizing co-expression analysis and its utility.
2. **Introduce the problem gently:** Use phrasing like "Despite its usefulness…" or "While co-expression analysis has become a standard tool…"
3. **Connect to your work:** Highlight that in the process of developing your software, a particular bias became apparent.
4. **Lead into the next section:** End with a sentence that signals you will explore this bias in detail.

Example transition sentence:
"While co-expression analysis provides valuable insights into gene regulatory networks, our experience developing [software tool name] revealed a recurring bias that can influence interpretation of the results. Understanding this bias is crucial for accurate downstream analyses and motivated the refinements incorporated in our tool."

If you want, I can draft a full paragraph that bridges your introduction to your discussion of the bias, keeping it concise and flowing naturally. Do you want me to do that?

You Asked

same paragraph?

ChatGPT gpt-5-mini

Yes — you can fit the transition in the same paragraph if you keep it concise. You'd start by summarizing co-expression analysis, then immediately introduce the bias you observed, and end with a hint that your tool addresses or investigates it.

Here's an example of a single paragraph version:

"Co-expression analysis is a widely used approach for identifying genes with coordinated expression patterns, providing insights into regulatory networks and functional relationships. While powerful, this method is not without limitations; during

the development of [software tool name], we observed a specific bias that can influence the interpretation of co-expression results. Recognizing and addressing this bias is essential for ensuring accurate analysis, and it motivated the design choices implemented in our tool."

This keeps the paragraph tight, smoothly moving from general background to the specific issue your work addresses.

If you like, I can also suggest a slightly more “storytelling” version that makes the transition feel even more natural. Do you want me to do that?